

Unconditional all-orders asymptotics for two OEIS permutation sequences

Stable path forests, Poisson expansions, and index inversion

Prepared for Vladimir Reshetnikov

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Abstract

The OEIS entries A189281 and A110128 count permutations avoiding a prescribed signed or absolute difference between entries two positions apart. Their displayed higher asymptotic expansions are associated with conjectured recurrences. We give a direct, recurrence-free proof of those expansions, together with a finite exact formula for every coefficient. More generally, for any fixed positive offsets r, s , the complete probability generating function of the number of violations has an all-orders expansion consisting of $e^{\theta u}$ times polynomial corrections, where $\theta = 1$ in the oriented case and $\theta = 2$ in the absolute case. The remainder is uniform on compact complex u -sets.

The structural ingredient is an exact stabilization theorem for tilings of long path forests: low factorial moments depend on the numbers of paths, not their individual lengths. We also prove a quantitative total-variation universality bound, a factorial denominator bound for all asymptotic coefficients, and an all-orders Lambert- W index inversion. Exact rational computations recover the OEIS coefficients through order ten and extend both expansions through order sixteen. Independent permutation enumeration and Bonferroni certificates accompany the proofs. The specific conjectured recurrences and the stronger all-orders integrality suggested by A189281 are not assumed or claimed proved. A concrete rational-function conjecture is proposed as a route to the latter question.

Status. This is a proof-focused research draft, not a claim of peer review or a Lean-checked formalization. The theorems below have mathematical proofs; the computations are reproducible exact checks. Historical priority beyond the sources reviewed here remains to be assessed. The results establish the algebraic asymptotic sector, not a complete exponentially improved transseries.

Editorial note (1 October 2026, batch 73O2). This report is manuscript 60 of batch 73 of ProveIt’s incoming-reports intake, placed in the research-report collection in commit [6e193dd4f](#). It is AI-assisted, unrefereed and not formalized. The text is the delivered manuscript; the write added only the label prefix `spf:`, the bracketed notes marked “[Added 1 October 2026, batch 73O2]”, Remarks [7.2](#) and [9.1](#), three bibliography entries, and [section B](#), which records the provenance and every editorial change. The rational-collapse statement of [section 11](#) remains a conjecture.

Editorial note (1 October 2026, batch 74). Since batch 74 the report has two Parts. Part I is manuscript 60, as described above. Part II ([section 14](#)) adds manuscript 03 of batch 74, an independent second derivation of Part I’s main theorems, and prints only what it adds: uniformity over forest shapes, comparison bounds, closed forms of $c_3(r, s)$, and proved integrality for the oriented one-path family. Its labels carry the prefix `fga:`.

Editorial note (3 October 2026, batch 85). Since batch 85 the report has three Parts. Part III ([section 23](#)) adds manuscript 03 of batch 85, which proves the rational collapse, Conjecture [11.1](#), for every h ([theorem 26.1](#)), extends it to every pair of directed offsets, and proves that every oriented correction polynomial $B_f^{(1)}(u; r, s)$ has integer coefficients, with an exact degree law ([theorem 28.1](#)). Its labels carry the prefix `spf:rc:`. The abstract, the batch-7302 note above, [section 11](#) and the conclusion of Part I describe manuscript 60 as delivered, in which the collapse is a conjecture and integrality is not claimed; dated notes at Conjecture [11.1](#), Proposition [11.2](#), questions 1 and 4 of [section 12](#), in Part II and in [section B](#) record what has changed. The conjectured recurrences of A189281 and A110128 are still not proved, and the absolute case is not covered by Part III.

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Part I

Unconditional all-orders asymptotics for two OEIS permutation sequences

1 The target and the precise contribution

For a uniformly random $\pi \in S_n$, define, for fixed $r, s \geq 1$,

$$X_{r,s,n}^{(1)} = \sum_{i=1}^{n-r} \mathbf{1}\{\pi(i+r) - \pi(i) = s\}, \quad (1.1)$$

$$X_{r,s,n}^{(2)} = \sum_{i=1}^{n-r} \mathbf{1}\{|\pi(i+r) - \pi(i)| = s\}. \quad (1.2)$$

An empty sum is zero. Write

$$A_{r,s}^{(\theta)}(n) = n! \mathbb{P}(X_{r,s,n}^{(\theta)} = 0), \quad \theta \in \{1, 2\}. \quad (1.3)$$

In this notation,

$$A189281(n) = A_{2,2}^{(1)}(n), \quad A110128(n) = A_{2,2}^{(2)}(n). \quad (1.4)$$

These identifications and the sequence definitions are recorded in the OEIS entries [1, 2].

1.1 What the sources do, and do not, already establish

The A189281 entry lists an order-eight, degree-eleven recurrence as conjectural, and describes its displayed terms through n^{-10} as consequences of that recurrence. It also records independent dynamic-programming verification of sequence values through $n = 300$ [1]. The recurrence-guessing work of Kauers and Koutschan is an important source for this problem [5]. For A110128, the entry lists a guessed order-24, degree-64 operator and attributes its higher asymptotic terms to that operator [2].

The common leading limits are not new. Tauraso proved, in the diagonal absolute case, the first correction

$$\frac{A_{r,r}^{(2)}(n)}{n!} = e^{-2} \left(1 + \frac{4(r-1)}{n} + O(n^{-2}) \right) \quad (1.5)$$

and gave exact counting formulas [4]. Spahn and Zeilberger developed an exact matching-of-tilings formula for both offset families, explained its computational use, and discussed Matsuo's alternative approach [3]. We use their combinatorial framework, with a self-contained proof, rather than claiming the underlying inclusion-exclusion formula as new.

There is also a relevant recent development. Padhi's August 2026 preprint reports holonomicity for every fixed r, s , while explicitly recording only a partial result toward the specific A189281 operator [6]. Our arguments neither assume that preprint's holonomicity theorem nor prove a particular recurrence. A general existence theorem for a recurrence and a proof of a specific guessed operator are distinct questions. [Added 1 October 2026, batch 73O2: the intake confirmed that the arXiv record of [6] exists under the title given in the bibliography, but did not check its content. Nothing below depends on it.]

1.2 The gap addressed here

We supply an unconditional route from the counting problem to every coefficient and a controlled remainder. The important point is not merely to fit more terms. One must justify replacing

an n -dependent inclusion-exclusion sum by an infinite coefficient calculation, including the contributions of arbitrarily many isolated selected edges. The exact stabilization theorem and a uniform factorial-moment bound supply that justification.

The resulting contribution has four connected parts. First, the coefficients attached to the two conjectural recurrences become theorems about the original permutation counts. Second, the same proof gives every fixed pair of offsets and the full violation distribution. Third, it explains why congruence classes of n do not enter any algebraic-order coefficient. Fourth, it gives an inverse expansion with a precise interpretation for a discrete sequence.

The ProveIt repository was consulted as the project context. Its Fabius asymptotic audit carefully separates a printed approximation, a necessary correction, and an all-orders theorem [9]. Here the analogous separation is between finite data, formal coefficient extraction, a uniform asymptotic theorem, and an unproved stronger closed form. No declaration from the repository is a mathematical dependency of the present proof, and no new repository build is reported.

1.3 A preview of the main theorem

Put

$$G_{r,s,n}^{(\theta)}(u) = \mathbb{E}(1+u)^{X_{r,s,n}^{(\theta)}}. \quad (1.6)$$

The use of $1+u$, rather than u , makes the coefficients factorial moments.

Theorem 1.1 (All-orders polynomial correction theorem). *For fixed positive integers r, s , $\theta \in \{1, 2\}$, every integer $M \geq 0$, and every $0 \leq U < \infty$, there are polynomials $B_J^{(\theta)}(u; r, s) \in \mathbb{Q}[u]$ such that, uniformly for $|u| \leq U$,*

$$G_{r,s,n}^{(\theta)}(u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s,U}(n^{-M-1}). \quad (1.7)$$

They are given by the finite formula (5.11). They satisfy

$$B_0 = 1, \quad \deg B_J \leq 2J, \quad B_J(0) = 0 \quad (J \geq 1), \quad (1.8)$$

$$B_J(u; r, s) = B_J(u; s, r), \quad (2J)!B_J(u; r, s) \in \mathbb{Z}[u], \quad (1.9)$$

$$B_1^{(\theta)}(u; r, s) = \theta(1-r-s)u + \theta(1-\theta)u^2. \quad (1.10)$$

In particular, for $c_J^{(\theta)}(r, s) = B_J^{(\theta)}(-1; r, s)$,

$$\frac{A_{r,s}^{(\theta)}(n)}{n!} = e^{-\theta} \left(\sum_{J=0}^M \frac{c_J^{(\theta)}(r, s)}{n^J} + O(n^{-M-1}) \right), \quad c_0 = 1, \quad c_1 = \theta(r+s-\theta). \quad (1.11)$$

The proof occupies sections 2 to 6. The arithmetic assertions are proved in section 7.

2 Selected violations and pairs of path tilings

2.1 Why paths are the right objects

The directed graph on $\{1, \dots, n\}$ with edges $i \rightarrow i+r$ is a disjoint union of r directed paths, when $n \geq r$. Their vertex counts are $\lfloor n/r \rfloor$ and $\lceil n/r \rceil$. The value graph with edges $j \rightarrow j+s$ is similarly a union of s paths.

More generally, consider any directed path forest on the position vertices and any directed path forest on the value vertices, both having n vertices. The oriented statistic counts position edges mapped to directed value edges by a uniform bijection. The absolute statistic counts position edges mapped to value edges in either direction. The proofs below apply to this more general model as well.

Selecting a subset of the position edges gives a collection of nontrivial path components and singleton vertices. For the oriented statistic, each selected component must map in its prescribed order to a value-path interval. For the absolute statistic there are exactly two choices of direction per nontrivial component. A reversal of direction *within* one component is impossible: at the first change of sign it would repeat the preceding value. Thus the orientation weight is 2^c , not 2^k , where c is the number of components and k the number of selected edges.

Definition 2.1 (Component profile). A profile is a finitely supported sequence $\mathbf{b} = (b_2, b_3, \dots)$ of nonnegative integers. Define

$$\begin{aligned} c &= \sum_{\ell \geq 2} b_\ell, & k &= \sum_{\ell \geq 2} (\ell - 1)b_\ell, \\ v &= \sum_{\ell \geq 2} \ell b_\ell = k + c, & d &= k - c = \sum_{\ell \geq 2} (\ell - 2)b_\ell, \end{aligned} \tag{2.1}$$

and $\mathbf{b}! = \prod_{\ell \geq 2} b_\ell!$. The number d is the *defect*: dimers have defect zero, and each extra edge in a component adds one unit of defect.

For a position forest $\mathcal{P}$, let $C_{\mathcal{P}}(\mathbf{b})$ be its number of tilings with b_ℓ components of size $\ell \geq 2$, and $n - v$ singletons. A tiling is simply a partition into consecutive intervals along each path; unused edges may lie between neighboring intervals.

Proposition 2.2 (Exact profile identity). *For position and value forests $\mathcal{P}, \mathcal{V}$ on n vertices,*

$$\mathbb{E}(1 + u)^X = \sum_{\mathbf{b}: v \leq n} \theta^c u^k C_{\mathcal{P}}(\mathbf{b}) C_{\mathcal{V}}(\mathbf{b}) \frac{(n - v)! \mathbf{b}!}{n!}. \tag{2.2}$$

Infeasible profiles have zero tiling count.

Proof. For each permutation, expand $(1 + u)^X$ as a sum over its selected violation sets. Such a set determines its input tiling and its image tiling, with the same component profile. Match the b_ℓ input components of each length to the b_ℓ output components in $b_\ell!$ ways. Choose a direction for each nontrivial component when $\theta = 2$, and match the $n - v$ singletons arbitrarily. This gives exactly $\theta^c (n - v)! \mathbf{b}!$ permutations for each pair of tilings, counted with the selected set distinguished. Division by $n!$ proves the identity.

Additional violations outside the selected set are allowed. This is essential: the identity counts marked subsets of violations, not permutations having exactly that selected set. $\square$

At $u = -1$, (2.2) is the matching-of-tilings inclusion-exclusion formula of Spahn and Zeilberger [3]. Its extension to general u makes the later distributional statements possible.

3 Exact stabilization on long path forests

Write $x^{\underline{j}} = x(x - 1) \cdots (x - j + 1)$ and $x^{\bar{j}} = x(x + 1) \cdots (x + j - 1)$, with both empty products equal to one. For a profile $\mathbf{b}$, put

$$E_h(\mathbf{b}) = [z^h] \prod_{\ell \geq 2} (1 + (\ell - 1)z)^{b_\ell}. \tag{3.1}$$

Define an integer polynomial

$$F_r(n; \mathbf{b}) = \sum_{h=0}^c (-1)^h (r - 1)^{\bar{h}} E_h(\mathbf{b}) (n - k - h)^{\underline{c-h}}. \tag{3.2}$$

Its degree in n is c and its leading coefficient is one. For $r = 1$, it reduces to $(n - k)^{\underline{c}}$.

Theorem 3.1 (Stable tiling polynomial). *Let $\mathcal{P}$ be a union of r directed paths with vertex counts $L_1, \dots, L_r$ and total n . If each $L_i \geq k$, then*

$$C_{\mathcal{P}}(\mathbf{b}) = \frac{F_r(n; \mathbf{b})}{\mathbf{b}!}. \quad (3.3)$$

Thus, in this range, the answer depends on $n, r, \mathbf{b}$, but not on the individual path lengths.

3.1 A formal coefficient identity

The following residue calculation is included to make the stabilization mechanism explicit. Only finitely many variables are needed for any specified coefficient.

Let $z_2, z_3, \dots$ be formal variables and let W , with zero constant term, solve

$$W = \sum_{\ell \geq 2} z_{\ell} (1 + W)^{-(\ell-1)}. \quad (3.4)$$

Define

$$\mathcal{D} = 1 + \sum_{\ell \geq 2} (\ell - 1) z_{\ell} (1 + W)^{-\ell}. \quad (3.5)$$

For every integer N and every profile,

$$[\mathbf{z}^{\mathbf{b}}] \frac{(1 + W)^N}{\mathcal{D}} = \frac{(N - k)^c}{\mathbf{b}!}. \quad (3.6)$$

To prove this, replace every z_{ℓ} by tz_{ℓ} and put $\phi(w) = \sum z_{\ell} (1 + w)^{-(\ell-1)}$. Then $W = t\phi(W)$ and $\mathcal{D} = 1 - t\phi'(W)$. Work first over the fraction field in finitely many z_{ℓ} , so that $\phi(0)$ is invertible; the resulting identity is polynomial coefficientwise. The change of formal residue variable $t = w/\phi(w)$ gives

$$[t^c] \frac{f(W)}{1 - t\phi'(W)} = [w^c] f(w) \phi(w)^c. \quad (3.7)$$

Indeed, the Jacobian factor cancels $1 - t\phi'(W)$. The coefficient of $\mathbf{z}^{\mathbf{b}}$ in $\phi(w)^c$ is $c!(1 + w)^{-k}/\mathbf{b}!$. Taking $f(w) = (1 + w)^N$ proves (3.6).

3.2 Proof of the stabilization theorem

For a single path of length L , a profile with $v \leq L$ has $L - k$ tiles in all. Ordering those tiles gives

$$C_L(\mathbf{b}) = \frac{(L - k)!}{(L - v)! \mathbf{b}!} = \frac{(L - k)^c}{\mathbf{b}!}. \quad (3.8)$$

If $k \leq L < v$, the falling factorial is zero, again giving the correct count. Consequently, through weighted edge degree $k \leq L$, the actual tiling series agrees with the formal surrogate $(1 + W)^L/\mathcal{D}$.

For r paths, all allocations of a total profile of edge degree k to the paths have local edge degree at most k . Under $L_i \geq k$, multiplying the surrogates is therefore legitimate at the required coefficient. Their product is

$$\frac{(1 + W)^n}{\mathcal{D}^r}. \quad (3.9)$$

Expand the additional factor $\mathcal{D}^{-(r-1)}$. A selected subprofile $\mathbf{a} \leq \mathbf{b}$, with $|\mathbf{a}| = h$, contributes

$$\frac{(-1)^h (r - 1)^{\bar{h}}}{\mathbf{a}!} \prod_{\ell \geq 2} (\ell - 1)^{a_{\ell}} \mathbf{z}^{\mathbf{a}} \frac{(1 + W)^{n - \sum \ell a_{\ell}}}{\mathcal{D}}. \quad (3.10)$$

Applying (3.6) to the remaining profile gives

$$\frac{(-1)^h (r-1)^{\bar{h}} \prod (\ell-1)^{a_\ell}}{\mathbf{a}!(\mathbf{b}-\mathbf{a})!} (n-k-h)^{\underline{c-h}}. \quad (3.11)$$

Summing over subprofiles of size h replaces the product of binomial coefficients by $E_h(\mathbf{b})/\mathbf{b}!$. This is exactly (3.2)–(3.3), proving theorem 3.1. $\square$

Corollary 3.2 (Stable factorial moments). *If the position forest has r paths, the value forest has s paths, every path has at least k vertices, and $2k \leq n$, then*

$$\mu_k := \mathbb{E} \binom{X}{k} = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{\theta^c F_r(n; \mathbf{b}) F_s(n; \mathbf{b})}{\mathbf{b}! n^v}. \quad (3.12)$$

For the ordinary offset model, this is exact whenever

$$k \leq \min \left(\left\lfloor \frac{n}{r} \right\rfloor, \left\lfloor \frac{n}{s} \right\rfloor, \left\lfloor \frac{n}{2} \right\rfloor \right). \quad (3.13)$$

Proof. Use theorem 3.1 on both tiling counts in (2.2) and extract u^k . The condition $2k \leq n$ ensures $v = k + c \leq n$ for every profile in the displayed sum. $\square$

As a first check, a single dimer has $F_r = n - r$, so, for $n \geq \max(r, s, 2)$,

$$\mathbb{E} X = \frac{\theta(n-r)(n-s)}{n(n-1)}. \quad (3.14)$$

The same formula follows directly by counting the available input and output edges.

4 Uniform bounds and beyond-all-orders universality

An exact formula for fixed moments is not, by itself, enough to justify an asymptotic expansion of a probability. A bound uniform in the moment order is needed.

Lemma 4.1 (Uniform factorial-moment bound). *For either statistic and arbitrary path forests on $n \geq 1$ vertices,*

$$0 \leq \mu_k \leq \frac{(\theta e^2)^k}{k!} \quad (k \geq 0). \quad (4.1)$$

Consequently, for $|u| \leq U$ and $a = \theta e^2 U$,

$$\left| G(u) - \sum_{k=0}^K \mu_k u^k \right| \leq e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.2)$$

Proof. Fix a selected set of k position edges, with c components and $v = k + c$ vertices. There are at most $\theta^c n^c$ ways to assign starts and directions to its components in the value forest, even before excluding overlaps. Hence the probability that all the selected edges are violations is at most $\theta^c n^c / n^v$.

For $0 \leq v \leq n$, the geometric mean of the largest v factors in $n!$ is at least the geometric mean of all n factors. Also $n! \geq (n/e)^n$. Thus $n^v \geq (n/e)^v$, and the last probability is at most $\theta^k e^{2k} n^{-k}$, since $c \leq k$ and $v \leq 2k$. There are at most $n^k/k!$ possible selected edge sets. Summing proves (4.1). Summing its exponential-series tail proves (4.2). For $k = 0$ the assertion is immediate. $\square$

Theorem 4.2 (Quantitative independence of long path lengths). *Consider two position-value forest pairs, each on n vertices, with the same numbers r and s of paths. Suppose every path in both pairs has at least K vertices and $2K \leq n$. Let X and $\tilde{X}$ be the corresponding statistics for the same θ . For every $R > 1$, with $a = \theta e^2(R + 1)$,*

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \frac{R}{R-1} e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.3)$$

If all path lengths are at least ηn for a fixed $\eta > 0$, the right-hand side can be chosen to be $\exp(-\eta' n \log n + O(n))$ for every fixed $0 < \eta' < \min(\eta, 1/2)$.

Proof. By Corollary 3.2, the two sets of factorial moments agree through order K . On the circle $|t| = R$, use $u = t - 1$ in (4.2); the two ordinary probability generating functions differ by at most $2e^a a^{K+1}/(K+1)!$. Cauchy's coefficient inequality bounds the difference of the probabilities at j by this quantity times R^{-j} . Sum over $j \geq 0$ and divide by two to obtain (4.3). Finally, take $K = \lfloor \eta' n \rfloor$ and use $\log(K!) = K \log K - K + O(\log K)$. $\square$

This statement is stronger than agreement of a few asymptotic coefficients. It concerns the entire probability law. It also clarifies a limitation: the bound does not identify the leading exponentially small difference between two particular forest geometries. That finer sector is left open.

5 A finite formula for every asymptotic coefficient

5.1 Normalization and the defect filtration

Put $x = 1/n$ and define

$$f_{r,\mathbf{b}}(x) = \sum_{h=0}^c (-1)^h (r-1)^{\bar{h}} E_h(\mathbf{b}) x^h \prod_{a=0}^{c-h-1} (1 - (k+h+a)x), \quad (5.1)$$

$$D_v(x) = \prod_{a=0}^{v-1} (1 - ax), \quad (5.2)$$

$$Q_{\mathbf{b}}(x) = \frac{f_{r,\mathbf{b}}(x) f_{s,\mathbf{b}}(x)}{D_v(x)}. \quad (5.3)$$

Then $f_{r,\mathbf{b}}(1/n) = n^{-c} F_r(n; \mathbf{b})$, and a stabilized profile contributes

$$\frac{\theta^c u^k}{\mathbf{b}!} n^{-d} Q_{\mathbf{b}}(1/n) \quad (5.4)$$

to the generating function. This displays why defect $d > J$ cannot contribute to the coefficient of n^{-J} .

Separate the dimers from the other components:

$$m = b_2, \quad \beta = (b_3, b_4, \dots). \quad (5.5)$$

For the latter profile write

$$C = \sum_{\ell \geq 3} \beta_\ell, \quad K = \sum_{\ell \geq 3} (\ell - 1) \beta_\ell, \quad d = K - C, \quad \beta! = \prod_{\ell \geq 3} \beta_\ell!. \quad (5.6)$$

Thus $c = m + C$, $k = m + K$, $v = 2m + K + C$. The profiles β of a given defect d are in bijection with ordinary integer partitions of d : a component of length ℓ contributes a part $\ell - 2$.

Lemma 5.1 (Polynomial dependence on the number of dimers). *For fixed β, j, r, s ,*

$$P_{\beta,j}(m) := [x^j]Q_{(m,\beta)}(x) \quad (5.7)$$

is a polynomial in m over $\mathbb{Q}$ of degree at most $2j$. For positive integer r, s , it takes integer values at every $m \geq 0$.

Proof. The coefficient $E_h(m, \beta)$ is a polynomial in m of degree at most h , since the dimer contribution is $(1+z)^m$. For a product of consecutive linear factors

$$\prod_{a=0}^{m+C-h-1} (1 - (m + K + h + a)x), \quad (5.8)$$

the coefficient of x^q is polynomial in m of degree at most $2q$. One way to see the bound is to take its formal logarithm. The coefficient of x^i is a constant multiple of a sum of i th powers over a consecutive interval with affine endpoints, and therefore has degree at most $i + 1$. Exponentiating shows degree at most $2q$, because a product of power sums of total x -degree q has degree at most $q + q$. The same argument applies to $D_v(x)^{-1}$.

In a term carrying the additional factor $E_h x^h$, the degree at coefficient x^j is at most $h + 2(j - h) \leq 2j$. Products preserve the total bound. For the few small nonnegative m for which $c < h$, $E_h = 0$; thus the polynomial extension gives the same zero contribution. Equivalently one can define the consecutive power sums by their polynomial extensions before multiplying by E_h .

Finally, for nonnegative m , the numerator polynomials in (5.3) have integer coefficients and $D_v^{-1} \in \mathbb{Z}[[x]]$. Hence every value in (5.7) is an integer. $\square$

5.2 Poisson summation by finite differences

Let $\Delta P(m) = P(m+1) - P(m)$. For a polynomial of degree at most q ,

$$P(m) = \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} m^{\underline{t}}. \quad (5.9)$$

Therefore the elementary identity

$$\sum_{m \geq 0} \frac{z^m}{m!} P(m) = e^z \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} z^t \quad (5.10)$$

turns an infinite dimer sum into a finite polynomial. It follows either by summing each falling factorial, or by differentiating e^z exactly t times.

Definition 5.2 (Coefficient formula). For $J \geq 0$, define

$$B_J^{(\theta)}(u; r, s) = \sum_{d=0}^J \sum_{\substack{\beta \geq 0 \\ \sum_{\ell \geq 3} (\ell-2)\beta_\ell = d}} \frac{\theta^C u^K}{\beta!} \sum_{t=0}^{2(J-d)} \frac{\Delta^t P_{\beta, J-d}(0)}{t!} (\theta u)^t. \quad (5.11)$$

All sums on the right are finite.

Only evaluations at $m = 0, 1, \dots, 2(J-d)$ are required. Their sufficiency is a proved degree bound, not a polynomial guess. No term $A(n)$ and no recurrence for $A(n)$ is needed to compute B_J .

6 Proof of the uniform all-orders expansion

We now justify the asymptotic interpretation of (5.11). This is the analytic step that a purely formal expansion of the profile formula would leave unresolved.

Lemma 6.1 (Uniform analytic bound for a profile). *For every profile of edge degree k , put $\rho_k = 1/[8(k+1)^2]$. There is a constant $C_{r,s}$, independent of the profile and k , such that*

$$\sup_{|x| \leq \rho_k} |Q_{\mathbf{b}}(x)| \leq C_{r,s}. \quad (6.1)$$

One may take $C_{r,s} = e^{2^{r+s}}$. In particular,

$$|[x^j]Q_{\mathbf{b}}(x)| \leq C_{r,s}[8(k+1)^2]^j, \quad (6.2)$$

and, if $n \geq 16(k+1)^2$,

$$\left| Q_{\mathbf{b}}(1/n) - \sum_{j=0}^q [x^j]Q_{\mathbf{b}}(x)n^{-j} \right| \leq 2C_{r,s} \left(\frac{8(k+1)^2}{n} \right)^{q+1}. \quad (6.3)$$

Proof. The elementary symmetric function bound $E_h(\mathbf{b}) \leq k^h/h!$ follows by expanding the h th power of the sum of the component edge lengths. Every integer appearing in a numerator factor is at most $2k$, and there are at most k such factors. Thus, on the stated disk,

$$|f_{r,\mathbf{b}}(x)| \leq e^{2k^2\rho_k} \sum_{h \geq 0} \frac{(r-1)^{\bar{h}}}{h!} (k\rho_k)^h = e^{2k^2\rho_k} (1 - k\rho_k)^{-(r-1)}. \quad (6.4)$$

For $r = 1$, the last sum is simply one. Also $v \leq 2k$ and $a\rho_k < 1/2$ for $0 \leq a < v$, so

$$|D_v(x)^{-1}| \leq \exp \left(2\rho_k \sum_{a=0}^{v-1} a \right) \leq e^{4k^2\rho_k}. \quad (6.5)$$

Combining these bounds gives (6.1) with the stated generous constant. The denominator has no zero on the disk. Cauchy's coefficient bound then proves (6.2), and its geometric tail at $1/n \leq \rho_k/2$ proves (6.3). $\square$

Proof of theorem 1.1: analytic assertion. Fix M, r, s, U , and for sufficiently large n choose

$$T_n = \left\lfloor \frac{(2M+4) \log n}{\log \log n} \right\rfloor. \quad (6.6)$$

The factorial-moment tail (4.2) with $K = T_n$ is $n^{-(2M+4)+o(1)}$, hence is $O(n^{-M-1})$. For fixed offsets, every position and value path has more than T_n vertices eventually. Also $2T_n \leq n$ and $16(T_n+1)^2 \leq n$. Thus all retained moments are given exactly by (3.12), and Lemma 6.1 applies to every retained profile.

High defect. The sum of absolute profile weights, marked by their defect, has the convergent generating function

$$\sum_{\mathbf{b}} \frac{\theta^c U^k}{\mathbf{b}!} z^d = \exp \left(\sum_{q \geq 1} \theta U^q z^{q-1} \right) = \exp \left(\frac{\theta U}{1 - Uz} \right). \quad (6.7)$$

For $U = 0$ the assertion is immediate; otherwise use a disk strictly inside $|z| < 1/U$. Its Taylor tail of degree M at $z = 1/n$ is $O(n^{-M-1})$. Together with (6.1), this bounds all retained profiles of defect $d \geq M+1$ by $O(n^{-M-1})$.

Low defect. For each $d \leq M$, there are only finitely many β . Expand $Q_{(m,\beta)}$ through order $M - d$. By (6.3), after multiplying by n^{-d} the error is bounded by n^{-M-1} times a fixed polynomial in m times $(\theta U)^m/m!$, with a constant depending only on the fixed parameters and β . The sum over m is finite uniformly in n . This is why no extra power of $\log n$ appears in the remainder.

We may now extend the retained dimer sums to $m = \infty$. Indeed, for each retained coefficient, (6.2) bounds the omitted summands by a polynomial in m times $(\theta U)^m/m!$. Since $k = m + K > T_n$ implies $m > T_n - K$, their tails are again $n^{-(2M+4)+o(1)}$. Summing each polynomial in m by (5.10) gives exactly (5.11). The total error is $O(n^{-M-1})$, uniformly on $|u| \leq U$. This proves (1.7) and, on setting $u = -1$, (1.11). $\square$

The proof also works for any fixed numbers r, s of paths whose shortest length is bounded below by a positive multiple of n . It does not assert uniformity for arbitrarily growing offsets or for forests with a bounded short path. [Added 1 October 2026, batch 74: Part II proves the expansion uniformly over forest shapes, on the disk $|1 + u| \leq 1$, whenever the shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7); remark 16.8 notes that the proof above already gives this on every disk $|u| \leq U$ with T_n in place of K_n .]

7 Arithmetic, first corrections, and the full distribution

7.1 Degree and denominator control

Proposition 7.1 (Arithmetic of the correction polynomials). *The structural assertions (1.9) hold.*

Proof. For a defect- d profile, $C \leq d$ and $K \leq 2d$. A term in (5.11) has u -degree at most $K + 2(J - d) \leq 2J$. The formula is symmetric in r, s because $Q_{\mathbf{b}}$ is. For $J = 0$ it gives $B_0 = 1$. When $J > 0$, any nonempty β contributes positive u -degree. For the empty profile the constant term is $P_{\emptyset, J}(0) = 0$, since the completely empty $\mathbf{b}$ has $Q_{\mathbf{b}} = 1$. Thus $B_J(0) = 0$.

By Lemma 5.1, every $\Delta^t P(0)$ is an integer. A summand's only possible denominator is $t!\beta!$. This divides $(t + C)!$, by the multinomial coefficient, and

$$t + C \leq 2(J - d) + d \leq 2J. \quad (7.1)$$

Hence $t!\beta!$ divides $(2J)!$. Multiplying the finite sum by $(2J)!$ therefore gives a polynomial over $\mathbb{Z}$. $\square$

This denominator bound is unconditional. It is deliberately weaker than claiming all the oriented coefficients themselves are integers; that stronger phenomenon is discussed in section 11.

7.2 The first two corrections

For the dimer-only profile,

$$P_{\emptyset, 1}(m) = -m^2 + (2 - r - s)m = -m^2 + (1 - r - s)m. \quad (7.2)$$

The only defect-one profile is a single component of length three, with $C = 1$, $K = 2$, and $P_{\beta, 0} = 1$. Applying (5.10) immediately gives (1.10). For $\theta = 1$, a calculation of the four profile types of defect at most two gives

$$B_2^{(1)}(u; r, s) = (r - 1)(s - 1)u + \binom{r + s - 1}{2}u^2. \quad (7.3)$$

For completeness, the required dimer-only polynomial is

$$P_{\emptyset,2}(m) = \frac{1}{2}m^4 + (r + s - 3)m^3 + \frac{r^2 + 2rs + s^2 - 5r - 5s + 7}{2}m^2 + \frac{-r^2 + r - s^2 + s}{2}m. \quad (7.4)$$

For one length-three component, $P_{\beta,1}(m) = -m^2 - (r + s)m + 3 - 2r - 2s$. The defect-two profiles are one length-four component or two length-three components; both have $P_{\beta,0} = 1$. These data and (5.11) verify (7.3) directly.

In the absolute case a useful general expression is

$$B_2^{(2)}(u; r, s) = 2(r - 1)(s - 1)u + 2((r + s)^2 - r - s - 1)u^2 + 2(2r + 2s - 1)u^3 + 2u^4. \quad (7.5)$$

Thus (1.5) is recovered as a special case of the first correction, while the theorem supplies every later order.

Remark 7.2 (Offsets (1,1): two classical sequences). [Added 1 October 2026, batch 73O2.] The manuscript does not discuss the smallest offsets, which give two classical sequences. The intake ran the shipped coefficient generator `code/path_forests.py` on a copy with `-r 1 -s 1 -order 10`.

- *Oriented case, no successions.* Here $A_{1,1}^{(1)}(n)$ counts permutations of $[n]$ with no $\pi(i + 1) = \pi(i) + 1$, which is $A000255(n - 1) = D_n + D_{n-1}$ with D_n the derangement numbers [10]. The formula gives $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots, 0$ for $J = 0, \dots, 10$. This is consistent with $(D_n + D_{n-1})/n! = e^{-1}(1 + 1/n) + O(1/n!)$, so every later coefficient is zero as well.
- *Absolute case, Hertzprung's problem.* Here $A_{1,1}^{(2)}(n)$ is A002464, the number of permutations with no rising or falling succession [11]. The formula gives

$$c_J^{(2)}(1, 1) = 1, 0, -2, -\frac{10}{3}, -6, -\frac{154}{15}, -\frac{88}{9}, \frac{5336}{105}, \frac{1612}{3}, \frac{2098234}{567}, \frac{36500686}{1575} \quad (J = 0, \dots, 10).$$

These are, term for term, the Abramson–Moser expansion displayed in the OEIS entry (computed by Kotesovec, 2011, extended 2020), through n^{-10} .

Both cases agree with $c_1 = \theta(r + s - \theta)$ of (1.11). As an independent check, the intake computed both sequences exactly to $n = 800$, from the recurrences printed in the two OEIS entries (themselves checked against brute-force enumeration for $n \leq 8$). It found the scaled residuals $n^{M+1}(e^\theta A(n)/n! - \sum_{j \leq M} c_j n^{-j})$ for $M = 4, 8$ approaching c_{M+1} at $n = 200, 400, 800$. These finite checks are evidence; the proof is [theorem 1.1](#). The theorem covers the pair (1,1), and the manuscript's brute-force tests already include it (it is one of the six offset pairs of [section 10](#)). Only the comparison with the two OEIS entries is new here.

7.3 Poisson corrections for all probabilities

The zero-violation event is only one coefficient of the probability generating function. Write $B_J(u) = \sum_h b_{J,h} u^h$ and define

$$H_h(q; \theta) = \sum_{i=0}^{\min(h,q)} \binom{h}{i} (-1)^{h-i} \frac{q^i}{\theta^i}. \quad (7.6)$$

These are a normalization of the Charlier correction polynomials. Direct coefficient extraction gives

$$[t^q](t - 1)^h e^{\theta(t-1)} = e^{-\theta} \frac{\theta^q}{q!} H_h(q; \theta). \quad (7.7)$$

Consequently define the signed approximation

$$p_M(q) = e^{-\theta} \frac{\theta^q}{q!} \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(q; \theta). \quad (7.8)$$

Corollary 7.3 (An all-orders approximation of the entire law). *For fixed M, r, s, θ ,*

$$\sum_{q \geq 0} \left| \mathbb{P}(X_{r,s,n}^{(\theta)} = q) - p_M(q) \right| = O(n^{-M-1}). \quad (7.9)$$

In particular the law converges in total variation to $\text{Poisson}(\theta)$, with error $O(n^{-1})$.

Proof. Apply [theorem 1.1](#) on $|t| = R > 1$, using $u = t - 1$. Cauchy's inequality gives an error $O(n^{-M-1}R^{-q})$ in the coefficient of t^q . Summing the geometric series proves (7.9). $\square$

For $M \geq 1$, p_M need not be nonnegative at every q , so it is a signed asymptotic approximation, not a new probability model. It nevertheless has total mass one, because $B_0(0) = 1$ and $B_J(0) = 0$ for $J > 0$.

Remark 7.4 (A clique-model analogue). [Added 1 October 2026, batch 73O2, reciprocal note.] The companion report [a330266-balanced-smirnov-poisson](#) of this collection (batch 73, manuscript 57) studies the number $X_{n,k}$ of equal-rank adjacencies in a uniform shuffle of kn cards, k in each of n ranks. There the value relation is n cliques K_k , not a fixed number of long paths, so the hypotheses of this report do not apply, and the limit is $\text{Poisson}(\lambda)$ with $\lambda = k - 1$. Its first-order local law (Corollary 8.3 there, label `bsw:cor:local`),

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right],$$

is the analogue of the case $M = 1$ of (7.8)–(7.9). That report proves it at first order only, and the analytic step behind its expansion is sketched; its Remark 6.3 names a uniform factorial-moment bound of the type of Lemma 4.1 as the device that would make it rigorous.

8 The two OEIS expansions, unconditionally

8.1 A189281: oriented distance-two avoidance

The first correction polynomials for $r = s = 2$, $\theta = 1$ are

$$B_0 = 1, \quad B_1 = -3u, \quad B_2 = u + 3u^2, \quad (8.1)$$

$$B_3 = u + u^2 - u^3, \quad B_4 = u - u^2 - 2u^3, \quad B_5 = u + u^2 - 3u^3, \quad (8.2)$$

$$B_6 = u + 23u^2 - 4u^3, \quad B_7 = u + 121u^2 + 19u^3. \quad (8.3)$$

Further examples are

$$B_8 = u + 479u^2 + 378u^3 + 24u^4, \quad (8.4)$$

$$B_9 = u + 1681u^2 + 3737u^3 + 672u^4 + 24u^5, \quad (8.5)$$

$$B_{10} = u + 5543u^2 + 28936u^3 + 11088u^4 + 984u^5 + 24u^6. \quad (8.6)$$

Setting $u = -1$ in these polynomials proves

$$\begin{aligned} \frac{\text{A189281}(n)}{n!} &= e^{-1} \left(1 + \frac{3}{n} + \frac{2}{n^2} + \frac{1}{n^3} + \frac{3}{n^5} + \frac{26}{n^6} + \frac{101}{n^7} + \frac{124}{n^8} \right. \\ &\quad \left. - \frac{1409}{n^9} - \frac{13266}{n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.7)$$

The coefficient of n^{-4} is exactly zero. The terms through n^{-10} agree with the recurrence-dependent list in the OEIS [1]; the proof above does not assume that recurrence.

8.2 A110128: absolute distance-two avoidance

For $r = s = 2$, $\theta = 2$, the first polynomials are

$$B_1 = -6u - 2u^2, \quad (8.8)$$

$$B_2 = 2u + 22u^2 + 14u^3 + 2u^4, \quad (8.9)$$

$$B_3 = 2u + 6u^2 - 60u^3 - 56u^4 - 16u^5 - \frac{4}{3}u^6, \quad (8.10)$$

$$B_4 = 2u - 18u^2 - 122u^3 + 74u^4 + 154u^5 + 70u^6 + 12u^7 + \frac{2}{3}u^8. \quad (8.11)$$

It follows that

$$\begin{aligned} \frac{\text{A110128}(n)}{n!} = e^{-2} & \left(1 + \frac{4}{n} + \frac{8}{n^2} + \frac{68}{3n^3} + \frac{242}{3n^4} + \frac{1692}{5n^5} \right. \\ & + \frac{72802}{45n^6} + \frac{2725708}{315n^7} + \frac{16083826}{315n^8} \\ & \left. + \frac{186091480}{567n^9} + \frac{32213578294}{14175n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.12)$$

This recovers the higher list attributed to the guessed operator in the OEIS [2], again by a direct argument from the permutation model. The fractions here also show why an integer-coefficient assertion cannot be transferred indiscriminately from the oriented case to the absolute case.

8.3 Additional coefficients and how to use them

J	$c_J^{(1)}(2, 2)$	$c_J^{(2)}(2, 2)$
11	-59103	2620164054016/155925
12	-9448	61699558213516/467775
13	2525459	6658719787519636/6081075
14	26632378	136086447719175784/14189175
15	152270749	5133067332126139204/58046625
16	106221948	546416517735316663856/638512875

Table 1: Exact evaluations of the finite coefficient formula beyond the order-ten lists. The accompanying JSON includes every polynomial B_J through order sixteen, not only its value at $u = -1$.

For any fixed M , adding terms through $c_M n^{-M}$ gives a remainder $O(n^{-M-1})$ inside the parentheses. This is a Poincaré statement with M fixed as $n \rightarrow \infty$. It does not establish convergence of the infinite series, justify setting M proportional to n , or identify an optimal truncation rule. The exact finite coefficients in [table 1](#) are supplied with their generating code rather than inferred from a numerical fit.

9 Lambert- W inversion with a discrete interpretation

Remark 9.1 (Relation to the repository’s inversion apparatus). [Added 1 October 2026, batch 7302.] This section re-derives, for its own forward map, machinery that the canonical volume *Transseries and Inversion* of the ProveIt repository already contains [12], and the manuscript does not cite it. The forward map $e^{-\theta} \Gamma(n+1) C(1/n)$ is the gamma carrier of that volume’s chapter on the increasing inverse of Gamma (theorem “All-orders expansion of Γ_+^{-1} ”, label `p6:thm:gamma`),

perturbed by a power series in $1/n$, which is the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`). The volume’s chapter “The subfactorial” (`p8:sec:top`) treats the closest analogue, the derangement numbers, by citing that carrier. Normalizations differ: here $z = L/W_0(L/e)$ with $L = \log y$, and the constant $\frac{1}{2} \log(2\pi)$ sits in κ_θ ; the volume uses $R = \log y - \frac{1}{2} \log(2\pi)$ and $T = R/W_0(R/e)$. The discussion of integer thresholds at the end of the section is an instance of the volume’s staircase theorem (`p0:thm:staircase`). [Theorems 9.2](#) and [9.3](#) are therefore printed as a self-contained second route for this family of sequences. No novelty is claimed for the inversion method; what is specific to this report is the forward input, the coefficients c_j of [theorem 1.1](#).

9.1 Logarithmic coefficients

For either sequence, or any fixed r, s, θ , suppress the parameters and write

$$\frac{A(n)}{n!} \sim e^{-\theta} C(1/n), \quad C(x) = 1 + \sum_{j \geq 1} c_j x^j. \quad (9.1)$$

The right side is a formal asymptotic series. Define $\log C(x) = \sum_{j \geq 1} \ell_j x^j$. An explicit coefficient recursion is

$$\ell_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k \ell_k c_{j-k}. \quad (9.2)$$

Combining the proved probability expansion with Stirling’s expansion, in its usual remainder-controlled form [7], yields

$$\log A(n) = n \log n - n + \frac{1}{2} \log n + \kappa_\theta + \sum_{j=1}^M a_j n^{-j} + O(n^{-M-1}), \quad (9.3)$$

where

$$\kappa_\theta = \frac{1}{2} \log(2\pi) - \theta, \quad a_j = \ell_j + \frac{B_{j+1}}{j(j+1)}. \quad (9.4)$$

Here B_m are Bernoulli numbers; the additional term vanishes for even j . For A189281,

$$\ell_1 = 3, \quad \ell_2 = -\frac{5}{2}, \quad \ell_3 = 4, \quad \ell_4 = -\frac{29}{4}, \quad a_1 = \frac{37}{12}, \quad a_2 = -\frac{5}{2}, \quad a_3 = \frac{1439}{360}. \quad (9.5)$$

For A110128, $a_1 = 49/12$.

9.2 The first explicit inverse terms

Given $y = A(n)$, let $L = \log y$ and

$$z = \frac{L}{W_0(L/e)}, \quad w = \log z. \quad (9.6)$$

The principal Lambert function is characterized by $W_0(x)e^{W_0(x)} = x$ for $x > 0$ [8]. Thus $z(\log z - 1) = L$ exactly.

Theorem 9.2 (First inverse corrections). *Along the sequence values $y = A(n)$ as $n \rightarrow \infty$,*

$$n = z - \frac{1}{2} - \frac{\kappa_\theta}{w} - \frac{1}{z} \left(\frac{c_1 - 1/24}{w} + \frac{\kappa_\theta^2}{2w^3} \right) + O\left(\frac{1}{z^2 w}\right). \quad (9.7)$$

For A189281 the coefficient $c_1 - 1/24$ is $71/24$; for A110128 it is $95/24$.

Proof. Set $n = z + D$. The derivative of $x \log x - x$ at z is w , and its second derivative is $1/z$. The constant-order terms of (9.3), after subtracting L , give $wD + \frac{1}{2}w + \kappa_\theta = 0$, hence $D = d_0 = -1/2 - \kappa_\theta/w$ at that order. The coefficient of z^{-1} is

$$wd_1 + \frac{d_0^2 + d_0}{2} + a_1. \quad (9.8)$$

Since $a_1 = c_1 + 1/12$, this gives

$$d_1 = -\frac{c_1 - 1/24}{w} - \frac{\kappa_\theta^2}{2w^3}. \quad (9.9)$$

The residual in the logarithmic equation is $O(z^{-2})$; division by a derivative asymptotic to w gives the asserted index error. A precise all-orders version of this argument follows next. $\square$

9.3 An all-orders inverse recursion

Put $t = 1/z$ and seek $D(t, w) = \sum_{j \geq 0} d_j(w)t^j$. Define

$$\begin{aligned} \mathcal{H}(t, D) = & \frac{(1 + tD) \log(1 + tD) - tD}{t} + \frac{1}{2} \log(1 + tD) \\ & + \sum_{j \geq 1} a_j t^j (1 + tD)^{-j}. \end{aligned} \quad (9.10)$$

The quotient by t is a formal series divisible by t , so no singular constant term occurs. The inverse equation is

$$wD + \frac{1}{2}w + \kappa_\theta + \mathcal{H}(t, D) = 0. \quad (9.11)$$

It gives the triangular recursion

$$d_0(w) = -\frac{1}{2} - \kappa_\theta/w, \quad (9.12)$$

$$d_j(w) = -\frac{1}{w} [t^j] \mathcal{H} \left(t, \sum_{q=0}^{j-1} d_q(w)t^q \right) \quad (j \geq 1). \quad (9.13)$$

Only $a_1, \dots, a_j$ enter the j th step.

Theorem 9.3 (All-orders index inversion). *For every fixed $J \geq 0$, the functions in (9.13) are rational in w , with coefficients in $\mathbb{Q}[\kappa_\theta]$ for the rational forward coefficients considered here, and*

$$n = z + \sum_{j=0}^J \frac{d_j(w)}{z^j} + O\left(\frac{1}{z^{J+1}w}\right), \quad y = A(n). \quad (9.14)$$

Moreover, $d_0 = O(1)$ and $d_j = O(1/w)$ for $j \geq 1$ as $w \rightarrow \infty$.

Proof. The recursion first proves the algebraic form of the coefficients by induction. Since $\mathcal{H}$ has no t^0 term, its coefficient of t^j depends only on the earlier d_q , and stays bounded when those earlier coefficients are bounded. Dividing by w proves the stated size bounds.

Fix J and retain sufficiently many terms of (9.3). Substitution of the truncated D cancels every term through z^{-J} , leaving a residual $O(z^{-J-1})$. All relevant derivatives of the finite logarithmic model are controlled on a bounded neighborhood of $z + d_0$, and its first derivative is $\log z + O(1/z)$. The actual index is in such a neighborhood: the leading term first gives $n/z \rightarrow 1$, and then the mean value theorem applied to $n \log n - n - z \log z + z = O(\log n)$ gives $n - z = O(1)$. A further application of the mean value theorem divides the residual by a quantity asymptotic to w , proving (9.14). $\square$

9.4 What is being inverted

A discrete sequence has no distinguished analytic interpolation. The preceding theorems make no such choice: their statements are evaluated at the actual values $y = A(n)$. Alternatively, define an explicit smooth approximation

$$\mathcal{A}_M(x) = e^{-\theta} \Gamma(x+1) \sum_{j=0}^M c_j x^{-j}. \quad (9.15)$$

It is positive and strictly increasing for all sufficiently large x , because the finite correction tends to one and the logarithmic derivative is $\log x + O(1/x)$. Its inverse satisfies

$$\mathcal{A}_M^{-1}(A(n)) - n = O\left(\frac{n^{-M-1}}{\log n}\right). \quad (9.16)$$

This is a well-defined model inverse, with its discrepancy from the sequence controlled.

For an integer threshold $N(y) = \min\{n : A(n) \geq y\}$, an asymptotic estimate alone does not justify unconditional nearest-integer rounding near a jump. Certified bounds for $A(n)$ can resolve the remaining adjacent candidates. The Bonferroni method in the next section provides one such tool. [Added 1 October 2026, batch 73O2: the general form of this remark, exact rounding $N(y) = \lceil \nu(y) \rceil$ for an increasing interpolation ν and the separation condition under which an approximate inverse decides it, is the staircase theorem `p0:thm:staircase` of [12]. Its separation step is formalized in Lean as `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. That formalization is context only; it does not verify any statement of this report.]

10 Reproducible computations and exact certificates

10.1 Coefficient generation

The core implementation, `code/path_forests.py`, uses only the Python standard library. It forms the numerator factors and reciprocal denominator in (5.3) as truncated integer polynomials; evaluates $P_{\beta,j}(m)$ at the proved sufficient set of integers; takes forward differences; and assembles (5.11) using exact rational arithmetic. The work depends on the requested asymptotic order, not on a large sequence index n .

At order J , profiles are indexed by partitions of $d \leq J$, and each profile uses at most $2(J-d)+1$ dimer evaluations. No integer-sequence terms are used as fitting data. The archive contains all B_J and c_J through order sixteen in `data/coefficients_order16.json`.

10.2 Independent finite checks

The included validation program completed the following checks.

Check	Exact equalities/cases
Full tiling enumeration against both OEIS lists, $n = 0, \dots, 21$	44
Brute-force full distributions for $n = 1, \dots, 8$, six offset pairs, and both statistics	96
Stable factorial moments against those brute-force distributions	246
Stable tiling polynomial on specified unequal path-length forests	86
Additional finite-difference tests beyond the interpolation degree	375
Factorial denominator checks on polynomial coefficients through order 16	396

The six offset pairs were $(1, 1), (1, 2), (2, 2), (2, 3), (3, 2), (3, 3)$. Full tiling enumeration is independent of the stable-polynomial implementation, and brute permutation enumeration is independent of both. These checks detect finite implementation and bookkeeping errors; none replaces the proofs for arbitrary n or arbitrary expansion order.

10.3 Bonferroni probability intervals

When (3.13) holds through $2q + 1$, exact factorial moments give

$$S_{2q+1} \leq \mathbb{P}(X = 0) \leq S_{2q}, \quad S_j = \sum_{k=0}^j (-1)^k \mu_k. \quad (10.1)$$

For completeness, this follows pointwise from $\sum_{k=0}^j (-1)^k \binom{x}{k} = (-1)^j \binom{x-1}{j}$ when $x \geq 1$, while both partial sums equal one for $x = 0$. Thus the lower and upper bounds are rigorous rational numbers, even though the full count has not been computed.

The certificate program uses S_{31} and S_{30} at $n = 80, 160, 320$. It brackets e^{-1} between the odd and even Taylor sums of degrees 101 and 102, and squares that interval for e^{-2} . All arithmetic in the resulting error intervals is rational. The quantity displayed below is

$$E_M(n) = n^{M+1} \left(e^{\theta} \frac{A_{2,2}^{(\theta)}(n)}{n!} - \sum_{j=0}^M c_j n^{-j} \right). \quad (10.2)$$

By the theorem its limit is c_{M+1} .

θ	n	Certified interval for $E_4(n)$	Certified interval for $E_8(n)$
1	80	[3.3409847, 3.3409848]	[-1584.0080, -1584.0079]
1	160	[3.1664733, 3.1664734]	[-1494.2195, -1494.2194]
1	320	[3.0822399, 3.0822400]	[-1451.0335, -1451.0334]
2	80	[360.08331, 360.08332]	[359523.99, 359524.00]
2	160	[348.86238, 348.86239]	[343097.56, 343097.57]
2	320	[343.54178, 343.54179]	[335473.66, 335473.68]

Table 2: Outward-rounded exact intervals. The limits are $3, -1409$ for $\theta = 1$, and $1692/5, 186091480/567$ for $\theta = 2$. The intervals certify these finite residuals; their convergence is a consequence of the proof, not an inference from three samples.

The archive includes the unrounded rational endpoints and additional order-ten residual certificates. With the same moment cutoff, those order-ten intervals become wider at larger n , especially for $\theta = 2$: the factor n^{11} magnifies the fixed Bonferroni width. This is an honest precision limit of that finite computation; increasing the cutoff addresses it without using a guessed recurrence.

10.4 Inverse diagnostics and reproducibility

Using exact enumerated sequence values at $n = 20$, the Lambert backbone z has index errors about 0.52023 for A189281 and 0.20749 for A110128. Including both displayed correction levels in (9.7) reduces these to approximately -0.00066084 and 0.00089419 , respectively. These are numerical diagnostics, not certified rounding guarantees.

The programs were run with Python 3.13.5. The core coefficient, enumeration, and rational-certificate programs need no third-party package. The optional rational-identity exploration uses SymPy 1.14.0; the inverse diagnostic uses mpmath 1.3.0. The commands and expected

outputs are recorded in the archive's README. No Lean code was executed or supplied as an asserted formal proof. [Added 1 October 2026, batch 73O2: the delivered README is not shipped. The report's README.md replaces it and gives rerun commands for the shipped layout; see [section B](#).]

11 A sharper conjecture suggested by the exact computations

There is a striking simplification in the oriented $(2, 2)$ case that is not needed by any preceding theorem. Define the universal rational moments

$$\mu_k^*(n) = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{F_2(n; \mathbf{b})^2}{\mathbf{b}! n^v}, \quad (11.1)$$

as rational functions in an indeterminate n . They agree with the actual moments whenever the stability conditions hold. Remove the leading Poisson factor by setting

$$R_h(n) = \sum_{k=0}^h \frac{(-1)^{h-k}}{(h-k)!} \mu_k^*(n). \quad (11.2)$$

Exact finite algebra gives

$$R_0 = 1, \quad (11.3)$$

$$R_1 = -\frac{3n-4}{n^2}, \quad (11.4)$$

$$R_2 = \frac{3n^2 - 17n + 26}{n^4}, \quad (11.5)$$

$$R_3 = -\frac{(n-6)(n^2 - 7n + 16)}{n^6}. \quad (11.6)$$

Conjecture 11.1 (Rational collapse). For every $h \geq 4$,

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^5}. \quad (11.7)$$

This has been checked as an exact identity of rational functions for each $h = 4, \dots, 16$. The check clears the common denominator n^{2h} and compares polynomials; it is stronger than testing finitely many numerical values of n , but it is still only a finite check in h . No induction or creative-telescoping certificate for all h has been obtained here. [Added 3 October 2026, batch 85: the conjecture is now proved for every $h \geq 4$, together with the ratio recurrence (11.8), in Part III ([theorem 26.1](#)). It is the case $r = s = 2$ of a closed form of R_h for every pair of directed offsets ([theorem 25.1](#)), proved by a single-sum formula for the stable moments and an alternating Vandermonde identity rather than by telescoping. The heading ‘‘Conjecture’’ and the label are kept so that references to this statement stay valid.]

The proposed formula would imply the recurrence

$$\frac{R_{h+1}(n)}{R_h(n)} = \frac{(n-2h+5)(n-2h+4)}{(n-h+1)(n-2h)(n-2h-1)} \quad (h \geq 4). \quad (11.8)$$

Proving (11.8) directly from (11.2) is therefore a sharply specified next target.

Proposition 11.2 (What the rational collapse would imply). *If Conjecture 11.1 holds, then $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$ for every J , and therefore all $c_J^{(1)}(2, 2)$ are integers. For $J \geq 4$ it would also give*

$$\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J-4). \quad (11.9)$$

Proof. The first four rational functions $R_h(1/x)$ have integer Taylor coefficients. Under the conjecture, each later one is $24x^{h+4}$ times a product of factors $(1 - ax)^{-1}$ with nonnegative integers a , after cancellation. It too has integer coefficients and begins at degree $h + 4$. At any fixed power x^J , only finitely many of these R_h can contribute.

By (11.2), the coefficient of u^h in the Poisson-normalized stable moment series is $R_h(n)$. Comparing with the already proved coefficient formula gives $B_J^{(1)}(u; 2, 2) = \sum_h [x^J] R_h(1/x) u^h$. The integrality and degree claims follow. $\square$

This conditional proposition must not be confused with the unconditional factorial denominator theorem. The conjecture packages the observed integrality into a more structured rational-function identity, making it more approachable than a bare statement about an infinite list of integers. [Added 3 October 2026, batch 85: since theorem 26.1 proves the hypothesis, the proposition now holds unconditionally (corollary 26.3). Part III also proves, by a second route, that $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for every pair of directed offsets, and that the degree bound is attained: $\deg B_J^{(1)}(u; 2, 2) = J - 4$, with leading coefficient 24, for every $J \geq 8$ (theorem 28.1). For the absolute case $\theta = 2$, which Part III does not cover, Proposition 7.1 remains the denominator statement.]

12 Further research questions and proposed next steps

1. Prove or refute the rational collapse. The most concrete next step is an identity in two symbolic parameters rather than a fit to sequence values. Clear denominators in (11.2), organize profiles by their component count, and seek a telescoping proof of (11.8). A combinatorial cancellation proof would be particularly valuable: it should explain both the factor 24 and the loss of four powers of n beyond the moment index. A failure at some later h would also be informative, because the first discrepancy would identify a missing component interaction. The all-orders integrality issue for the oriented family remains separate until such a proof exists. [Added 1 October 2026, batch 74: for the oriented one-path subfamily, $r = 1$ or $s = 1$, all-orders integrality is proved in Part II (corollary 18.2); the case (2, 2) of A189281 remains open.] [Added 3 October 2026, batch 85: the rational collapse is proved in Part III (theorem 26.1), by a single-sum formula for the stable moments (theorem 24.1) and Vandermonde convolutions, not by telescoping. The factor 24 is $4!$, from a component-index shift of four, and the four lost powers of n are the vanishing binomial coefficients with $j < r + s$ (section 25). All-orders integrality now holds for every oriented pair (r, s) , A189281 included (theorem 28.1). A combinatorial cancellation proof, also asked for here, is still open (section 34, question 9).]

2. Connect the asymptotic theorem to the specific guessed operators. A proved asymptotic expansion does not establish the A189281 or A110128 recurrence. Conversely, a rigorous operator certificate could supply a second, independent route to large-order coefficient behavior. One useful project is to translate the stable moment framework into a finite-dimensional holonomic representation with boundary terms retained, then compare its annihilator with the listed operators. All boundary terms must be proved, not discarded because their numerical influence is small.

3. Identify the first geometry-dependent exponential sector. Theorem 4.2 shows that long path-length differences are invisible to every algebraic order. It does not give their leading amplitude, sign, or possible dependence on n modulo the offsets. The first profiles that can see a short boundary have edge degree comparable to the shortest path. A saddle-point analysis of that moving profile boundary could reveal the first exponentially small sector, and would turn the present algebraic expansion into part of a genuine transseries description.

4. Determine large-order growth and justified optimal truncation. The coefficients through order sixteen already show that sign and magnitude can change markedly. The present proof allows constants to depend on M . It would be useful to prove explicit growth bounds in M , decide whether a Gevrey-type estimate is sharp, and derive a valid truncation rule for a given n . The rational-collapse conjecture, if proved, would make the oriented case a tractable testing ground, but the absolute case needs its own analysis. [Added 3 October 2026, batch 85: for the oriented case Part III proves that the Borel transform of the coefficients is entire (theorem 30.1), so that $|c_J| \leq \gamma_A A^{J-1} (J-1)!$ for every $A > 0$. Sharp growth, a summation method and an optimal truncation rule remain open (section 34, questions 3 and 4).]

5. Allow offsets or path counts to grow with n . The current proof uses fixed r, s , both in the shortest-path condition and in the analytic bound on $Q_{\mathbf{b}}$. When r or s grows, the shortest path may meet the moment cutoff, and the number of path boundaries changes the first corrections. Determine the largest growth range in which a useful uniform expansion persists, and the transition law when path lengths become comparable to $\log n / \log \log n$. This question requires new uniform estimates; it is not an immediate corollary of the fixed-offset theorem. [Added 1 October 2026, batch 74: for fixed r, s , Part II shows that the expansion survives whenever the shortest path is long compared with $\log n / \log \log n$ (corollary 16.7); the transition at that scale remains open (section 22, question 2).]

6. Extend from paths to several forbidden offsets and to cycles. For several allowed differences, selected components can branch or contain cycles. Their embeddings are no longer determined by one starting value and a single orientation. A useful extension would classify component types by an appropriate defect and prove an analogue of the uniform moment bound and finite coefficient formula. Cyclic boundaries are another natural test: short cycles can contribute at a finite algebraic order, whereas long cycles may only change a beyond-all-orders sector.

7. Formalize the proof and build a certified coefficient library. The finite profile identity, the stable tiling polynomial, and the integer coefficient arithmetic are natural first formalization targets. The analytic layer then needs the uniform factorial-moment tail, Cauchy bounds, and finite asymptotic truncation. A robust implementation should expose the exact stability hypotheses instead of silently applying F_r outside its range. The rational Bonferroni certificates are suitable artifacts for an independent checker. This is a proposed formalization route, not a report of completed Lean verification.

8. Combine asymptotic inversion with exact threshold certification. For very large y , the Lambert backbone and a few correction terms should reduce the inverse problem to a small number of candidate indices. Use exact Bonferroni intervals to compare each candidate count with y , adaptively increasing the moment cutoff only when necessary. The interesting algorithmic question is a proved cost bound that accounts for near-threshold inputs, where asymptotic rounding alone cannot decide the answer.

13 Conclusion

The main result is a direct all-orders theorem for a natural pair of permutation sequences whose higher displayed OEIS expansions were tied to guessed recurrences. Its proof does not solve those recurrence problems by assumption: it bypasses them. The stable path-tiling identity gives exact low moments, the factorial tail controls the remaining moments, and finite differences

sum the unlimited dimer contributions. Together they turn a formal asymptotic calculation into a uniform theorem.

The same structure yields more than two lists of numbers: a full-law Poisson correction expansion, a quantitative explanation of boundary-length universality, an arithmetic denominator theorem, and a rigorous inverse construction. The stronger rational-collapse pattern is separated from these proved results and posed as a specific next problem. That distinction is central to the usefulness of the report: the established theorems can be used now, while the remaining conjectures have explicit, testable targets.

Part II

Fixed-gap asymptotics: a second derivation and its own results

14 Provenance, scope and notation of Part II

[Added 1 October 2026, batch 74.] Part II was added in the write of batch 74. It is built from manuscript 03 of that batch and prints only what that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, and manuscript 03’s route is recorded in [section 15](#). Everything in Part II is written in Part I’s notation ([table 4](#)).

14.1 Provenance

Manuscript 03 of batch 74 is the archive `ProveIt_Fixed_Gap_Asymptotics.zip`, which arrived in commit `c664fc2f0` and was placed in this report by commit `b669cff87`. Its article is titled *All-Orders Asymptotics and Boundary Universality for Fixed-Gap Permutations* (25 pages, dated October 1, 2026); its title page reads “Prepared for Vladimir Reshetnikov” and its PDF metadata names “ChatGPT, research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `291fbe6bb4477e0cb2131f1e041b12a9abe363fd` (2026-10-01, 18:10; [\[14\]](#)).

It was written without sight of manuscript 60, the source of Part I, and proves Part I’s main theorems in independent text: its structure, notation, constants and several proof devices differ throughout, and the two texts share about 1.1% of their word 8-grams, almost all of them bibliography entries and the digits of the coefficient list. Every number the two manuscripts share agrees exactly: the coefficients of A_{189281} and A_{110128} through order twelve, the general second correction in r, s for both orientations, the polynomials B_1, B_2, B_3 at $(r, s) = (2, 2)$, the logarithmic coefficients and the first inverse coefficients.

[Added 1 October 2026, batch 74: manuscript 03 states that “a repository code search for A_{189281} returned no match at the time of the audit”, and its delivered `02-fixed-gap-sources.md` repeats it. The pinned commit `291fbe6bb` already held the archive of manuscript 60, `docs/incoming/oeis_path_forests.zip` (arrival commit `aa43cc555`), which code search cannot read; that archive was placed an hour later as Part I of this report (commit `6e193dd4f`). Part II is an independent derivation of the same theorems plus the results listed below.]

Scope and non-claims of manuscript 03, kept. Its contribution is “the explicit stable-profile and finite-defect framework, the uniform error analysis, and the quantitative comparison of whole laws”. The leading Poisson behaviour, Tauraso’s first correction and the matching-of-tilings inclusion–exclusion of Spahn and Zeilberger are credited as prior work. The proofs

are ordinary mathematics, not a Lean-certified development, and no publication priority is claimed. “All orders” means a remainder theorem at each fixed order, not a convergent or resurgent completion; no Borel summability, optimal truncation, Stokes constant or complete family of exponentially small sectors is asserted. The conjectured OEIS recurrences of A189281 and A110128 are not proved, and neither is all-orders integrality when both component counts exceed one. Its numerical tables are not interval certificates, and its values of A189281 at $n = 80, 120$ are b-file inputs, not regenerated. It made no OEIS submission and modified no repository file.

14.2 What Part II adds, and how it compares with Part I

Topic	Part I (manuscript 60)	Part II (manuscript 03)
All-orders remainder	fixed offsets r, s ; uniform on every disk $ u \leq U$ (theorem 1.1)	only on $ 1 + u \leq 1$, i.e. $ u \leq 2$, but uniform over all forest shapes whose shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7)
Coefficients	B_J through order 16, $(2J)!B_J \in \mathbb{Z}[u]$ (Proposition 7.1)	same values through order 12 (not new); B_J polynomial in r, s of total degree at most J (proposition 17.1); closed forms of $c_3^{(\theta)}(r, s)$ (proposition 17.2)
Integrality	for $(2, 2)$ oriented only conditional on Conjecture 11.1	<i>proved</i> for the oriented case with $r = 1$ or $s = 1$ (corollary 18.2); $(2, 2)$ untouched
Universality	total variation bound (theorem 4.2)	bounds for the whole generating function and the avoidance probability, a second total-variation bound, and an $n = 12$ example (theorem 16.3 and remark 16.4)
Whole law	ℓ^1 approximation of the law (Corollary 7.3)	explicit first-order local laws at $(2, 2)$ (19.2)–(19.3)
Inverse	all orders in the coordinates z, w (theorem 9.3)	explicit terms through N^{-3} in a coordinate absorbing κ_θ ; a_2, a_3 for A110128 (proposition 20.1)
Certificates	certified Bonferroni intervals (table 2)	uncertified decimals only (section 21)

Table 3: Part II against Part I. The remainder theorem of Part II is weaker in u and stronger in geometry; see also [remark 16.8](#).

14.3 Notation

Manuscript 03 and Part I use several letters for different objects, and manuscript 03 overloads D, w, T, A and B internally. [Table 4](#) fixes the convention: Part II uses Part I’s names for every shared object and new names ($K_n, D_{n,k}, \Lambda, N, \Delta, \omega_j, \varepsilon_M, \alpha$) for the rest. No normalization is changed.

15 Manuscript 03’s second routes to Part I’s theorems

[Table 5](#) records, for each result of Part I that manuscript 03 proves again, the device its proof uses. None of these statements is reprinted. Where a device is also needed for a new result of

Notion	Part I and Part II	Manuscript 03
orientation weight (1 or 2)	θ	η (Part I's η is a length proportion)
length proportion	α (Part II); η in theorem 4.2	α
generating function $\mathbb{E}\binom{X}{k}$	$G(1+u) = \mathbb{E}(1+u)^X$ μ_k	$G(w)$, $w = 1+z$, $z = u$ S_k (Part I's S_j are Bonferroni sums)
profile, multiplicity factorial	$\mathbf{b} = (b_\ell)$, $\mathbf{b}!$	$\mathbf{m} = (m_j)$, $\mathbf{m}!$
tiling count	$C_{\mathcal{P}}(\mathbf{b})$	$T_{\mathcal{F}}(\mathbf{m})$
stable tiling polynomial	$F_r(n; \mathbf{b})$	$\mathbf{b}!^{-1} F_r$ is its $T_r(n; \mathbf{m})$
formal root	$1+W$, $\mathcal{D}^{-1}$	λ , A
dimers; long profile	m ; β , C , K	h ; $\mathbf{m}$, c_0 , k_0
normalized profile kernel	$Q_{\mathbf{b}}(x)$	$H_h(t)$ (<i>not</i> Part I's H_h)
Charlier polynomials	$H_h(q; \theta)$	$Q_m(j; \eta)$ (<i>not</i> Part I's $Q_{\mathbf{b}}$)
correction polynomials	$B_J^{(\theta)}(u; r, s)$	$C_J(r, s; \eta, z)$
avoidance coefficients	$c_J^{(\theta)}(r, s)$	$c_J^{(\eta)}(r, s)$ (same)
moment cutoff $(1 - 2k/n)^{-2k}$	T_n (Part I), K_n (Part II) $D_{n,k}$	K (Part I's K is an edge degree) $D_{n,k}$ (its D is also $\log N$)
logarithmic coefficients	a_j	d_j (Part I's d_j are inverse coefficients)
$\frac{1}{2} \log(2\pi) - \theta$	κ_θ	γ
inverse coordinates	Λ , N , $\Delta = \log N$ (Part II)	L , N , D
inverse coefficients	$\omega_j(\Delta)$	$u_j(D)$
unscaled truncation error	$\varepsilon_M(n)$	$E_M(n)$ (Part I's E_M is scaled by n^{M+1})

Table 4: Notation of manuscript 03 against Part I. Part II uses the middle column throughout.

Part II, it is stated there.

Part I	Manuscript 03's route
Proposition 2.2	the same matching-of-tilings argument, stated for binomial moments μ_k , with the same orientation lemma (weight θ^c , not θ^k); adds the symmetry $A_{r,s} = A_{s,r}$ via $\pi \mapsto \pi^{-1}$
(3.6)	a formal residue identity for the root $\lambda = 1 + \sum_{\ell} z_{\ell} \lambda^{1-\ell}$, with $A = (1 + \sum_{\ell} (\ell-1) z_{\ell} \lambda^{-\ell})^{-1}$; here $\lambda = 1 + W$ and $A = \mathcal{D}^{-1}$
theorem 3.1	the identical polynomial, written $T_r(n; \mathbf{m}) = F_r(n; \mathbf{b})/\mathbf{b}!$
Lemma 4.1	$\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$ (lemma 16.1): sharper for $k \ll n$, Part I's holds for every k
(4.2)	a Taylor remainder on the unit disk $ 1+u \leq 1$ (lemma 16.2), not an exponential-series tail
theorem 4.2	total variation through Parseval and Cauchy–Schwarz (theorem 16.3); the $\exp(-\delta n \log n + O(n))$ form is Part I's last assertion with η' renamed δ
Definition 5.2	the same finite formula; the dimer sum is evaluated by Touchard polynomials, $e^{-x} \sum_m x^m m^q / m! = T_q(x)$ with $T_{q+1} = x(T_q + T'_q)$, instead of finite differences, and the denominator by exponentials of power sums
Proposition 7.1	$B_0 = 1$, $\deg_u B_J \leq 2J$, $B_J(0) = 0$ and symmetry; adds the degree in r, s (proposition 17.1); Part I's denominator theorem is not in manuscript 03
(1.10), (7.3), (7.5)	B_1 by the same two-profile hand calculation; general B_2 and B_3 in symbolic r, s in <code>data/02-fixed-gap-general_pgf_first_three.txt</code> ; B_2 agrees with Part I identically in r, s, u
theorem 1.1	theorem 16.6: a defect-sensitive moment bound and a kernel bound on $ x \leq 1/(D(1+k)^2)$ in place of Lemma 6.1
(8.7), (8.12), table 1	the coefficients of both sequences through order 12, equal to Part I's (see below)
residue classes (section 1)	the same explanation by theorem 3.1
(3.14)	the same count; adds $\mathbb{E}X = \theta + \theta(1-r-s)/n + \theta(r-1)(s-1) \sum_{q \geq 2} n^{-q}$
(9.3), theorems 9.2 and 9.3	the same logarithmic series; the inverse in a coordinate absorbing κ_{θ} (proposition 20.1)
section 9, last subsection	the same smooth carrier $\mathcal{A}_M$ and threshold bracket

Table 5: Results of Part I proved again by manuscript 03, and the device of each second route.

Order twelve is not new. Manuscript 03 presents its coefficients through order twelve as an extension of the OEIS lists (“extend them through order twelve”) and proposes appending

$$-\frac{59103}{n^{11}} - \frac{9448}{n^{12}} \text{ to A189281, } \frac{2620164054016}{155925 n^{11}} + \frac{61699558213516}{467775 n^{12}} \text{ to A110128.}$$

[Added 1 October 2026, batch 74: Part I's table 1 already gives orders 11 to 16 of both sequences. Manuscript 03's values at orders 0 to 12, computed independently by its own engine, agree with Part I's exactly; they confirm Part I's table and are not new.]

The inverse is not new as a method. As for Part I (Remark 9.1), manuscript 03's inverse is an instance of the transseries volume's apparatus [12]: its coordinates are those of `p6:thm:gamma` with $\log Y$ shifted by θ , its inverse expansion is a perturbed inversion around that gamma

carrier (`p0:thm:perturbed-inversion`), and its threshold bracket is `p0:thm:staircase`. No novelty is claimed for the method.

16 Uniformity over forest shapes

Throughout this section $\mathcal{P}$ and $\mathcal{V}$ are arbitrary directed path forests on n vertices with r and s paths, as in [section 2](#), and

$$D_{n,k} = \left(1 - \frac{2k}{n}\right)^{-2k} \quad (1 \leq k < n/2). \quad (16.1)$$

Lemma 16.1 (A sharper binomial-moment bound; manuscript 03). *For either statistic and arbitrary path forests, $\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$.*

Proof (manuscript 03). As in [Lemma 4.1](#), a selected set of k position edges with c components has probability at most $\theta^c n^c / n^{k+c}$. Since $c \leq k$ and $k + c \leq 2k$, this is at most $\theta^k n^{-k} D_{n,k}$; there are at most $n^k/k!$ such sets. $\square$

Lemma 16.2 (Taylor remainder on the unit disk; manuscript 03). *For $|1 + u| \leq 1$ and every integer $K \geq 1$,*

$$\left| G(1 + u) - \sum_{k=0}^{K-1} \mu_k u^k \right| \leq |u|^K \mu_K. \quad (16.2)$$

Proof (manuscript 03). For an integer $x \geq K$, Taylor's integral remainder for w^x about $w = 1$ along the segment from 1 to $w = 1 + u$, which stays in the closed unit disk, has modulus at most $\binom{x}{K} |u|^K$; for $x < K$ it vanishes. Take expectations. $\square$

Theorem 16.3 (Quantitative boundary universality; manuscript 03). *Let $(\mathcal{P}, \mathcal{V})$ and $(\tilde{\mathcal{P}}, \tilde{\mathcal{V}})$ be two pairs of forests on n vertices with r position and s value paths, every path of all four having at least K vertices, $1 \leq K < n/2$, and let $X, \tilde{X}$ be their statistics for the same θ . Then $\mu_k = \tilde{\mu}_k$ for $0 \leq k \leq K$, and*

$$\sup_{|w| \leq 1} |G(w) - \tilde{G}(w)| \leq \frac{2(2\theta)^K}{K!} D_{n,K}, \quad (16.3)$$

$$|\mathbb{P}(X = 0) - \mathbb{P}(\tilde{X} = 0)| \leq \frac{2\theta^K}{K!} D_{n,K}, \quad (16.4)$$

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \sqrt{n+1} \frac{(2\theta)^K}{K!} D_{n,K}. \quad (16.5)$$

If every path has at least αn vertices, all three are at most $\exp\{-\delta n \log n + O_{\alpha, \delta, \theta}(n)\}$ for every fixed $0 < \delta < \min(\alpha, 1/2)$.

Proof (manuscript 03). Moments of order $k \leq K$ are stable in all four forests and agree by [Corollary 3.2](#). Subtract the common Taylor polynomial of degree $K - 1$ and apply [lemmas 16.1](#) and [16.2](#) to both remainders, with $|u| = |w - 1| \leq 2$ on the disk and $|u| = 1$ at $w = 0$. For the total variation, write $G - \tilde{G} = \sum_{j \leq n} q_j w^j$; Parseval on the unit circle gives $\sum |q_j|^2 \leq \sup_{|w|=1} |G - \tilde{G}|^2$, and Cauchy-Schwarz gives $\frac{1}{2} \sum |q_j| \leq \frac{1}{2} \sqrt{n+1} \sup |G - \tilde{G}|$. For the last assertion take $K = \lfloor \delta n \rfloor$: $\log D_{n,K} = O(n)$ and $\log K! = \delta n \log n + O(n)$. $\square$

The bounds [\(16.3\)–\(16.4\)](#) on the whole generating function and on the avoidance probability are new here; [\(16.5\)](#) and the $\exp(-\delta n \log n)$ form are a second route to [theorem 4.2](#). Boundary placement can change the exact count, but, when all components are long, its normalized influence is smaller than every power of $1/n$; the theorem compares shape-dependent corrections and does not identify an exponentially small expansion for a single sequence.

Remark 16.4 (Agreement of moments is not equality of laws; manuscript 03). At $n = 12$ compare the position and value path lengths $(6, 6)/(6, 6)$ with $(5, 7)/(4, 8)$. Their moments agree through $k = 4$ and first differ at $k = 5$. The oriented avoidance counts are 222822578 and 222822556, the absolute ones 91310696 and 91284560, by manuscript 03's independent exact tiling routine. "All algebraic orders agree" must not be read as "the finite distributions are identical".

Lemma 16.5 (A defect-sensitive moment bound; manuscript 03). *Let $\mu_{k,c}$ be the part of μ_k from selected sets with exactly c components, so $d = k - c$. For $1 \leq c \leq k < n/2$,*

$$\mu_{k,c} \leq \frac{\theta^c}{c!} \binom{k-1}{c-1} n^{-d} D_{n,k} \leq \frac{\theta^k}{k!} \left(\frac{k^2}{n}\right)^d D_{n,k}. \quad (16.6)$$

Proof (manuscript 03). There are at most $\frac{n^c}{c!} \binom{k-1}{c-1}$ selected sets with c components and k edges: choose the c initial vertices, order them, and give them a composition of k into c positive edge counts (overlaps are overcounted). Each has probability at most $\theta^c n^c / n^{k+c} \leq \theta^c n^{-k} D_{n,k}$, because $n^{k+c} \geq n^{k+c} (1 - 2k/n)^{2k}$ as in lemma 16.1. Multiplying, and using $n^c \cdot n^{-k} = n^{-d}$, gives the first bound. For the second, $\binom{k-1}{c-1} = \binom{k-1}{d} \leq k^d$, $k!/c! \leq k^d$ and $\theta \geq 1$. $\square$

Theorem 16.6 (All orders, uniformly over forest shapes; manuscript 03). *Fix $r, s \geq 1$, $\theta \in \{1, 2\}$ and $M \geq 0$, and for large n put*

$$K_n = \left\lfloor \frac{(M+4) \log n}{\log \log n} \right\rfloor. \quad (16.7)$$

For every pair of path forests on n vertices with r and s paths, each path having at least K_n vertices,

$$G(1+u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s}(n^{-M-1}), \quad |1+u| \leq 1, \quad (16.8)$$

with B_J the polynomials of (5.11). The implied constant does not depend on the individual path lengths.

Proof (manuscript 03, condensed). Tail. By lemmas 16.1 and 16.2 with $|u| \leq 2$, truncating at $k < K_n$ costs at most $(2\theta)^{K_n} D_{n,K_n} / K_n!$. Since $K_n^2/n \rightarrow 0$, $D_{n,K_n} = \exp(O(K_n^2/n))$, and $\log K_n! = (M+4 + o(1)) \log n$; so the cost is $o(n^{-M-1})$. Every retained profile has $k < K_n$ edges and is stable in both forests (Corollary 3.2).

High defect. By lemma 16.5, with $k^2/n < 1/2$ for $k < K_n$, the retained profiles of defect $d \geq M+1$ contribute at most $O(n^{-M-1}) \sum_{k \geq 1} (2\theta)^k k^{2M+2}/k! = O_M(n^{-M-1})$, uniformly in u and in the path lengths.

Low defect. For each of the finitely many long profiles β of defect $d \leq M$, expand the normalized kernel $Q_{(m,\beta)}(1/n)$ of (5.3) through order $M-d$. Manuscript 03 proves that the kernel is analytic and bounded by a constant depending only on r, s on $|x| \leq 1/(D_0(1+k)^2)$ for a suitable D_0 , which is the role of Lemma 6.1; Cauchy's estimate gives a remainder $O((1+k)^{2(M-d)+2} n^{-(M-d)-1})$, summable against the weights $(\theta|u|)^m/m!$ of the dimers.

Completion. Extending each retained dimer sum from $m+K < K_n$ to all $m \geq 0$ adds tails of order $\exp(-K_n \log K_n + O(K_n)) = o(n^{-M-1})$. The completed sums are the Poisson transforms (5.10) of the polynomials (5.7), and collecting the powers of $1/n$ gives (5.11). No step uses the path lengths beyond the stability of the retained profiles. $\square$

Corollary 16.7 (How long is long enough; manuscript 03). *For fixed r, s , the expansion (16.8) holds to every fixed order for every family of forests whose shortest path length $L_{\min}(n)$ satisfies*

$$\frac{L_{\min}(n) \log \log n}{\log n} \rightarrow \infty. \quad (16.9)$$

In particular the paths need not have lengths proportional to n .

Proof (manuscript 03). For each fixed M , (16.9) eventually gives $L_{\min}(n) \geq K_n$. $\square$

Remark 16.8 (Part I's proof gives the same uniformity on every disk). [Added 1 October 2026, batch 74; this observation is the write's, not either manuscript's.] The remark after the proof of [theorem 1.1](#) claims the theorem for forests whose shortest path is a positive multiple of n . Its proof uses the path lengths only to know that every retained profile, of edge degree at most $T_n = \lfloor (2M + 4) \log n / \log \log n \rfloor$, is stable: the moment tail (4.2), the kernel bound Lemma 6.1 and the defect majorant (6.7) hold for arbitrary forests. Read so, it gives [theorem 1.1](#) uniformly on $|u| \leq U$ for all forests whose shortest path has at least T_n vertices, which also covers the condition (16.9). Manuscript 03 states and proves the shape-uniform form, on the unit disk only.

The theorem gives a remainder at each fixed order; it does not assert that the series in $1/n$ converges, and its proof provides no small optimized numerical constant for a given finite n .

17 The correction polynomials as polynomials in the offsets

Proposition 17.1 (Degree in the offsets; manuscript 03). *For every J and θ there is a polynomial in $\mathbb{Q}[r, s, u]$ of total degree at most J in (r, s) whose values at positive integers r, s are $B_J^{(\theta)}(u; r, s)$; it is symmetric in r, s and has degree at most $2J$ in u .*

Proof (manuscript 03, in Part I's notation). In (5.3), the offset r enters $f_{r,\mathbf{b}}(x)$ only through the rising factorials $(r-1)^{\overline{h}}$, of degree h in r , and each is accompanied by x^h ; so $[x^j]f_{r,\mathbf{b}}$ has degree at most j in r , and likewise for s . Hence $P_{\beta,j} = [x^j]Q_{(m,\beta)}$ has total degree at most j in (r, s) , and (5.11) uses $P_{\beta,J-d}$ with $d \geq 0$. The other assertions are Proposition 7.1. $\square$

Proposition 17.2 (The third avoidance coefficient; manuscript 03). *For all positive integers r, s ,*

$$\begin{aligned} c_1^{(1)} &= r + s - 1, & c_1^{(2)} &= 2(r + s - 2), \\ c_2^{(1)} &= \frac{1}{2}(r^2 - r + s^2 - s), & c_2^{(2)} &= 2(r^2 + rs - 2r + s^2 - 2s), \\ c_3^{(1)} &= \frac{1}{6}(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6), & & (17.1) \\ c_3^{(2)} &= \frac{2}{3}(2r^3 + 3r^2 + 9rs - 17r + 2s^3 + 3s^2 - 17s + 10). \end{aligned}$$

The first four are Part I's (1.11), (7.3) and (7.5) at $u = -1$; the closed forms of $c_3^{(\theta)}(r, s)$ are new. Manuscript 03 obtains them by evaluating its finite formula symbolically in r, s (the full B_3 for both orientations is in `data/02-fixed-gap-general_pgf_first_three.txt`); this write checked both against Part I's generator at $(r, s) = (1, 1), (1, 3), (2, 3), (3, 3), (2, 5)$. At $r = s = d$, $c_1^{(2)} = 4(d-1)$ is Tauraso's correction (1.5).

18 The oriented one-path family: proved integrality

Proposition 18.1 (Rational collapse for one path; manuscript 03). *Let the position forest be a single path ($r = 1$), the value forest have s paths, and $\theta = 1$. Then*

$$n! \mu_k = \binom{n-s}{k} (n-k)! \quad (0 \leq k \leq n-s), \quad (18.1)$$

so that $G(1+u) = \frac{1}{n^s} \sum_{k=0}^{n-s} \frac{u^k}{k!} (n-k)^s$, and, for $|u| \leq 2$ and a suitable constant C_s ,

$$G(1+u) = \frac{e^u}{n^s} \sum_{t \geq 0} \frac{\Delta^t p_n(0)}{t!} u^t + O(\exp\{-n \log n + C_s n\}), \quad p_n(m) = (n-m)^s. \quad (18.2)$$

For $s = 1$ the rational factor is $1 - u/n$; for $s = 2$ it is $1 - 2u/n + u^2/(n(n-1))$.

Proof (manuscript 03). Work on the value side. Any k of its $n - s$ edges contract the value set to $n - k$ blocks, and these blocks can be laid along the position path in $(n - k)!$ ways, with no further restriction; this is (18.1). Then use $(n - s)^k/n^k = (n - k)^s/n^s$. Completing the sum to all $k \geq 0$ adds only zero terms through $k = n$ and a factorial tail bounded by a degree- s polynomial times $2^k/k!$, divided by n^s ; the completed sum is e^u times the finite-difference Poisson transform (5.10) of p_n , a polynomial of degree s in m . $\square$

Corollary 18.2 (An integrality subfamily; manuscript 03). *In the oriented case, if $r = 1$ or $s = 1$, then $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for every J , and every avoidance coefficient $c_J^{(1)}(r, s)$ is an integer.*

Proof (manuscript 03). The finite sum in (18.2) is $\sum_q a_q(n) T_q(u)$, where $p_n(m) = \sum_q a_q(n) m^q$ has integer coefficients and $T_q(u) = e^{-u} \sum_{m \geq 0} u^m m^q / m!$ are the Touchard polynomials, which have integer coefficients; so it is an integer polynomial in n and u of degree s in n . Dividing by $n^s = n^s \prod_{i < s} (1 - i/n)$ and expanding $\prod_{i < s} (1 - i/n)^{-1}$, whose power-series coefficients are integers, gives an expansion in $1/n$ with coefficients in $\mathbb{Z}[u]$. By uniqueness of asymptotic coefficients it is $\sum_J B_J^{(1)}(u; 1, s) n^{-J}$. The case $s = 1$ follows by the symmetry $B_J(u; r, s) = B_J(u; s, r)$. $\square$

This proves integrality only for the oriented one-path subfamily. It says nothing about the case (2, 2) of A189281, whose all-orders integrality remains conditional on Conjecture 11.1 (Proposition 11.2); manuscript 03 likewise warns that the subfamily does not justify extrapolating convergence, termination or integrality to two paths on both sides. At $(r, s) = (1, 1)$ the factor $1 - u/n$ gives $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots$, the values found in Remark 7.2. [Added 3 October 2026, batch 85: the case (2, 2), and every other pair of directed offsets, is now proved in Part III (theorem 28.1), of which corollary 18.2 is the case $r = 1$ or $s = 1$; the conditional proposition cited above holds unconditionally (corollary 26.3). Proposition 18.1 is likewise the case $r = 1$ of theorems 24.1 and 25.1, which manuscript 03 of batch 85 derives without citing it (remark 25.2).]

19 Fixed local probabilities

For fixed j and M , coefficient extraction from (1.7) on a fixed circle gives

$$\mathbb{P}(X = j) = e^{-\theta} \frac{\theta^j}{j!} \left\{ \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(j; \theta) + O_j(n^{-M-1}) \right\}, \quad (19.1)$$

with $B_J = \sum_h b_{J,h} u^h$ and the Charlier polynomials (7.6). This is the pointwise form of Corollary 7.3, which is the stronger statement (it controls the summed ℓ^1 error); it asserts no relative-error expansion uniformly for j growing with n . Manuscript 03 makes the first order explicit for the two sequences: from $B_1^{(1)} = -3u$ and $B_1^{(2)} = -6u - 2u^2$ at $(r, s) = (2, 2)$,

$$\mathbb{P}(X_{2,2,n}^{(1)} = j) = \frac{e^{-1}}{j!} \left\{ 1 + \frac{3(1-j)}{n} + O_j(n^{-2}) \right\}, \quad (19.2)$$

$$\mathbb{P}(X_{2,2,n}^{(2)} = j) = e^{-2} \frac{2^j}{j!} \left\{ 1 + \frac{4-j(j+1)/2}{n} + O_j(n^{-2}) \right\}. \quad (19.3)$$

At $j = 0$ these are the first avoidance corrections; at $j = 1, 2$ they show how probability mass is redistributed, which an avoidance count alone does not.

20 The inverse in a coordinate absorbing the constant

For A189281 and A110128 the logarithmic coefficients (9.4) are

Sequence	a_1	a_2	a_3
A189281	37/12	-5/2	1439/360
A110128	49/12	0	4319/360

The first row and a_1 for A110128 are in section 9; a_2 and a_3 for A110128 are added by manuscript 03.

Manuscript 03 inverts (9.3) in coordinates that absorb κ_θ . For large Y put

$$\Lambda = \log Y - \kappa_\theta, \quad N = \frac{\Lambda}{W_0(\Lambda/e)} = \exp(W_0(\Lambda/e) + 1), \quad \Delta = \log N, \quad (20.1)$$

so that $N(\log N - 1) = \Lambda$ exactly. Part I's coordinates (9.6) are the same construction with $\log Y$ in place of Λ ; so $N(Y)$ is Part I's z evaluated at $e^{-\kappa_\theta} Y$.

Proposition 20.1 (First inverse corrections; manuscript 03). *The formal inverse $n_*(Y)$ of (9.3) satisfies*

$$n_*(Y) = N - \frac{1}{2} + \frac{1/8 - a_1}{N\Delta} + \frac{1/24 - a_1/2 - a_2}{N^2\Delta} + O\left(\frac{1}{N^3\Delta}\right), \quad (20.2)$$

and to every fixed order $n_*(Y) = N - \frac{1}{2} + \sum_{j \leq J} \omega_j(\Delta) N^{-j} + O(N^{-J-1}\Delta^{-1})$, with each ω_j rational in Δ and computable from $a_1, \dots, a_j$; for instance

$$\omega_3(\Delta) = -\frac{1}{\Delta} \left\{ \frac{(1/8 - a_1)^2}{\Delta} + \frac{(1/8 - a_1)^2}{2\Delta^2} + \frac{a_1}{4} + a_2 + a_3 - \frac{1}{64} \right\}. \quad (20.3)$$

For the two sequences,

$$n_*(Y) = N - \frac{1}{2} - \frac{71}{24N\Delta} + \frac{1}{N^2\Delta} + \dots, \quad n_*(Y) = N - \frac{1}{2} - \frac{95}{24N\Delta} - \frac{2}{N^2\Delta} + \dots, \quad (20.4)$$

with κ_1 and κ_2 respectively in (20.1).

Proof (manuscript 03). Put $n = N + \delta$. The dominant part $x \log x - x$ has derivative Δ and second derivative $1/N$ at N ; after subtracting Λ the equation becomes $0 = \Delta\delta + \frac{\delta^2}{2N} - \frac{\delta^3}{6N^2} + \dots + \frac{\Delta}{2} + \frac{1}{2} \log(1 + \delta/N) + \sum_j a_j(N + \delta)^{-j}$. The constant term gives $\delta = -\frac{1}{2} + O(1/(N\Delta))$; with $\delta = -\frac{1}{2} + \sum_j \omega_j N^{-j}$ the next two orders read $\Delta\omega_1 + a_1 - \frac{1}{8} = 0$ and $\Delta\omega_2 + \frac{1}{2}a_1 + a_2 - \frac{1}{24} = 0$. At each later order the new coefficient enters only as $\Delta\omega_j$. For a fixed truncation the uncanceled residual is $O(N^{-J-1})$ and the derivative of the smooth carrier is $\Delta + O(1/N)$, so the mean value theorem divides it by a quantity asymptotic to Δ . $\square$

The first coefficient agrees with Part I: $1/8 - a_1 = 1/24 - c_1$ is the coefficient $-(c_1 - 1/24)$ of theorem 9.2, which in Part I's coordinates also carries the terms in κ_θ that are absorbed here. Manuscript 03 interprets n_* as the inverse of the same smooth carrier as Part I, with $X_M(A(n)) = n + O(n^{-M-1}/\log n)$, and brackets the integer threshold $\nu(Y) = \min\{n : A(n) \geq Y\}$ between $[X_M(Y) \mp \epsilon_Y]$, $\epsilon_Y = O(N^{-M-1}/\log N)$, with an existence constant only; this bracket is `p0:thm:staircase` of [12]. It notes that expanding W_0 in logarithms would give the familiar nested logarithmic transseries in Y , and that Λ depends on θ , so one Λ must not be reused for both sequences.

21 Manuscript 03's computations

Manuscript 03's `verify.py` (shipped as `code/02-fixed-gap-verify.py`) checks, with assertions: 90 avoidance counts and 90 complete histograms against brute-force permutations for $n = 0, \dots, 8$ and the offset pairs $(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)$, both orientations; its exact tiling enumerator against A189281 for $n = 0, \dots, 26$ and $n = 40$; 132 stable-profile identities and a nonrealizable zero case; the moment agreement and disagreement of [remark 16.4](#); 50 one-path identities ([18.1](#)); all posted coefficients of both sequences through order ten, and orders 11 and 12; the general first correction and the mean through order three; and that the residual of ([20.4](#)) vanishes through N^{-3} . Its exact enumerator and its coefficient engine are independent of each other and of Part I's programs. At order twelve the engine enumerates the 77 long profiles of defect twelve, 272 of defect at most twelve, with the number of dimers summed exactly.

n	$\varepsilon_2(n)$	$\varepsilon_5(n)$	$\varepsilon_{10}(n)$	$\varepsilon_{12}(n)$
20	$1.2642 \cdot 10^{-4}$	$4.8572 \cdot 10^{-7}$	$-2.3683 \cdot 10^{-10}$	$5.4066 \cdot 10^{-11}$
40	$1.5661 \cdot 10^{-5}$	$6.9763 \cdot 10^{-9}$	$-1.3658 \cdot 10^{-13}$	$4.8917 \cdot 10^{-15}$
80	$1.9541 \cdot 10^{-6}$	$1.0406 \cdot 10^{-10}$	$-6.8418 \cdot 10^{-17}$	$5.2425 \cdot 10^{-19}$
120	$5.7883 \cdot 10^{-7}$	$8.9918 \cdot 10^{-12}$	$-7.9394 \cdot 10^{-19}$	$2.5777 \cdot 10^{-21}$

Table 6: Manuscript 03: the unscaled errors $\varepsilon_M(n) = e \text{A189281}(n)/n! - \sum_{j \leq M} c_j^{(1)}(2, 2)n^{-j}$, at 110-digit working precision, rounded. The values A189281(80) and A189281(120) are b-file inputs [[13](#)], not regenerated. These are uncertified decimals; Part I's [table 2](#) gives certified intervals for the scaled errors at $n = 80, 160, 320$.

[Added 1 October 2026, batch 74: scaled by n^{M+1} , the last two columns are consistent with Part I's higher coefficients: $n^{11}\varepsilon_{10}(n) = -58771$ and -58990 at $n = 80, 120$, against $c_{11}^{(1)}(2, 2) = -59103$; $n^{13}\varepsilon_{12}(n) = 2.882 \cdot 10^6$ and $2.758 \cdot 10^6$, approaching $c_{13}^{(1)}(2, 2) = 2525459$ from above as the next term $c_{14}^{(1)}/n$ with $c_{14}^{(1)} = 26632378$ predicts $(2525459 + 26632378/80 \approx 2.858 \cdot 10^6)$.] The four-term inverse ([20.4](#)) at $Y = \text{A189281}(n)$ misses n by about $2.0566 \cdot 10^{-4}$, $1.9555 \cdot 10^{-5}$, $1.9282 \cdot 10^{-6}$ and $5.0462 \cdot 10^{-7}$ at $n = 20, 40, 80, 120$; the signs are not predicted by the theorem. Manuscript 03 records its run as “Completed in 22.44 seconds” and its README speaks of “tens of seconds”. On this repository's machine the order-twelve run took more than 170 seconds at the intake and 104 seconds at the write, and an order-nine run 72 seconds. The run time depends on the machine and its load.

22 Further questions raised by manuscript 03

Manuscript 03 poses twelve research questions, as proposals only. Six overlap [section 12](#): a certificate for the particular recurrence (Part I's question 2); integrality beyond the one-path case, which asks whether $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for all r, s and so whether all $c_J^{(1)}(2, 2)$ are integers (question 1, and [corollary 18.2](#) for what is proved; [added 3 October 2026, batch 85:] both now hold, by [theorem 28.1](#)); large-order growth (question 4); growing gaps and component counts (question 5); cycles and other sparse constraint graphs (question 6); and a small formal package, starting from the profile identity, the stable polynomial, a verified coefficient checker and the universality bound (question 7). The other six are new here.

1. *A sharper oriented degree bound.* Does $\deg_u B_J^{(1)}(u; r, s) \leq J$ hold at every order, sharpening $2J$? The explicit polynomials through order three show the cancellation; such a bound would constrain the factorial cumulants of X . [Added 3 October 2026, batch 85: yes, for every r, s . Part III proves $\deg_u B_J^{(1)}(u; r, s) \leq \min\{J, \max(r + s - 1, J - r - s)\}$, with equality $J - r - s$ for $J \geq 2(r + s)$ when $r, s \geq 2$ ([theorem 28.1](#)).]

2. *Sharp short-path thresholds.* Condition (16.9) is sufficient, not shown optimal. What is the first shape-dependent term when a shortest path has fixed length, or grows like $c \log n / \log \log n$?
3. *Optimal quantitative universality.* Find the correct constant in the $n \log n$ exponent of theorem 16.3 for given proportions, and the path balances that maximize the difference.
4. *Several gaps simultaneously.* The joint law of violations for several position/value gap pairs, with interacting edge colours and components containing cycles; a first target is two pairs sharing one gap.
5. *Growing local counts and moderate deviations.* Extend (19.1) to a quantified range $j = j(n) \rightarrow \infty$.
6. *Faster generation and certified errors.* Reduce the cost of many orders through profile symmetries, falling-factorial bases or direct Touchard operations, and extract explicit constants from theorem 16.6 to obtain a certified finite-input threshold algorithm (compare Part I's question 8).

OEIS and formalization, as manuscript 03 proposes them. It suggests an OEIS addition stating that the expansions through n^{-10} have a recurrence-independent proof, appending orders 11 and 12, and keeping the recurrences' conjectural status; it submitted nothing, and Part I already has orders 11 to 16. It proposes a formalization in three layers (finite combinatorics; algebra of the stable polynomial and coefficient generation; analysis), with a coefficient checker that verifies equality with the finite formula rather than with a guessed recurrence. It supplies no Lean source, and nothing in Part II is formalized.

The shipped files of manuscript 03. The ten files of manuscript 03 carry the prefix 02-fixed-gap-: its two scripts in `code/`, its six recorded outputs and `requirements.txt` in `data/`, and `sources.md` at the report root, all byte-identical to the delivery. `verify.py` imports its sibling by the delivered name (`from fixed_gap import ...`) and writes unprefixes files into `data/` beside its own directory; run from `code/` it would fail to import, and if made to run it would write beside Part I's files under unprefixes names. The report README gives a copy-based recipe.

Part III

Rational collapse and universal integrality for directed offsets

23 Provenance, scope and notation of Part III

[Added 3 October 2026, batch 85.] Part III was added in the write of batch 85. It is built from manuscript 03 of that batch, which proves Conjecture 11.1 of Part I for every h , extends the identity behind it to every pair of directed offsets (r, s) , and derives from it that every oriented correction polynomial $B_j^{(1)}(u; r, s)$ has integer coefficients, with an exact degree law. Where manuscript 03 proves a statement already printed in Part I or Part II, the statement is not reprinted: its route is recorded in table 9, and its special cases that repeat Part II's propositions 17.2 and 18.1 are printed as notes crediting Part II (remarks 25.2 and 28.2). Everything in Part III is written in Part I's notation (table 8).

23.1 Provenance

Manuscript 03 of batch 85 is the archive `OEIS_Rational_Collapse_and_Integrality.zip` (16 files, one of them the checksum ledger `MANIFEST.sha256`, which lists the other 15), which arrived in commit `317c1ce2e` and was placed in this report by commit `713149ded`. Its article is titled *Rational collapse and universal integrality for directed permutation avoidance: Resolving the A189281 path-forest conjecture and extending it to every fixed pair of offsets* (23 pages, dated 3 October 2026); its title page reads “Prepared for Vladimir Reshetnikov, AI-assisted mathematical research report”, and its PDF metadata names “AI-assisted research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `d68b65ea064084fb121df9bb431b21d086f7be81` (2026-10-03, 17:08; [19]). It read this report’s `article.tex` as the Git blob `5b60a1ad447afe7dce58534fba447b05e401f172`, which is also the blob of this file before the batch-85 write: manuscript 03 saw Parts I and II exactly as they stood.

Two different manuscripts numbered 03 have contributed to this report. In Part III, “manuscript 03” means manuscript 03 of batch 85; in Part II it means manuscript 03 of batch 74, the source of Part II, and Part III always names that one with its batch.

Manuscript 03 states correctly what Parts I and II contain: the stable tiling theorem and the all-orders expansion, Conjecture 11.1 checked for $h = 4, \dots, 16$, its conditional integrality consequence (Proposition 11.2), and Part II’s integrality for the oriented one-path family. It does not cite Part II’s proposition 18.1 or proposition 17.2, which it re-derives as special cases; this write credits them.

Scope and non-claims of manuscript 03, kept. It is an unrefereed, proof-focused, AI-assisted research draft, not a Lean-checked formalization, and it makes no claim of exhaustive historical priority; its firm novelty comparison is with the explicitly open conjecture of this report. It does *not* prove, or assume, the order-eight, degree-eleven recurrence listed in the OEIS for the finite sequence A189281: the recurrences it proves are in the moment order h , (11.8), and for auxiliary inverse-factorial coefficients, (29.8), not in n . Its integrality theorem is for directed (oriented) differences only; the absolute model ($\theta = 2$, A110128) is not covered, and integrality must not be transplanted to it. Stable rational moments equal actual moments only in the stability range (3.13). Its hypergeometric and inverse-factorial series are formal and are not asserted to converge; its entire Borel transform does not establish Laplace summability or convergence of $\sum_J c_J n^{-J}$, and no beyond-all-orders term is identified, since forests with equal stable moments can differ. No sign rule for the c_J is claimed. Its index reconstruction numerics are uncertified, and no explicit onset for rounding is given. Its certificate limits follow from the theorem, not from extrapolation. Missing proofs on an OEIS page are not evidence that no proof exists, and its OEIS note is an unsubmitted draft, not an accepted contribution. Finite computations support its proofs but are not their logical basis, and no Lean kernel run is asserted.

23.2 What Part III adds

Source and novelty ledger of manuscript 03, restated. The OEIS sequence definitions and their displayed initial asymptotics are external, and no authorship is claimed for them; the A189281 entry, inspected on 3 October 2026, displays coefficients through n^{-10} and notes their integrality, an observation that is not by itself a theorem about every order. The paired path tilings are credited to Spahn and Zeilberger [3]. The stable tiling polynomial and the earlier all-orders expansion are credited to Part I; manuscript 03 reconstructs the stable identity with proof and gives a new proof of the analytic transfer through the collapse. Manuscript 03 claims as its own: the single-sum moment formula, the universal normalized identity and the A189281 product, integrality at every order with the sharp degree law, the inverse-factorial kernel and

Topic	Parts I and II	Part III (manuscript 03 of batch 85)
Rational collapse at $(2, 2)$	Conjecture 11.1, checked for $h = 4, \dots, 16$	<i>proved</i> for every $h \geq 4$, with (11.8) (theorem 26.1)
Stable moments	a sum over all profiles of edge degree k (3.12)	a single sum of $k + 1$ terms, every (r, s) (theorem 24.1)
Normalized moments R_h	$(2, 2)$ only, explicit for $h \leq 3$ (section 11)	a closed form for every (r, s) and h (theorem 25.1)
Integrality, $\theta = 1$	$(2J)!B_J \in \mathbb{Z}[u]$ (Proposition 7.1); $B_J \in \mathbb{Z}[u]$ if $r = 1$ or $s = 1$ (corollary 18.2); $(2, 2)$ conditional (Proposition 11.2)	$B_J \in \mathbb{Z}[u]$ for every (r, s) , by an explicit Stirling-number formula (theorem 28.1); Proposition 11.2 unconditional (corollary 26.3)
Degree in u , $\theta = 1$	$\leq 2J$; $\leq \max(3, J - 4)$ at $(2, 2)$ under the conjecture; Part II asks whether $\leq J$ (section 22, question 1)	$\leq \min\{J, \max(\sigma - 1, J - \sigma)\}$, $\sigma = r + s$; equal to $J - \sigma$, with leading coefficient w_σ , for $J \geq 2\sigma$ when $r, s \geq 2$
Coefficient growth	none proved (Part I, question 4)	entire Borel transform; $ c_J \leq \gamma_A A^{J-1} (J-1)!$ for every $A > 0$ (theorem 30.1)
Coefficients	$c_J(2, 2)$ for $J \leq 16$ (table 1); $c_3^{(\theta)}(r, s)$ (proposition 17.2)	$c_J(2, 2)$ for $J \leq 100$ (data), printed to $J = 24$; A189282–A189284 through c_6 (table 10)
Inverse	all orders (theorem 9.3); two terms in Part II's coordinates (proposition 20.1)	the same two terms, the second in closed form $(r-1)(s-1)$ for every pair (proposition 32.1)
Certificates	$n = 80, 160, 320$, $\theta = 1, 2$ (table 2)	$n = 64, 128, 256$, $\theta = 1$, from single-sum moments (table 11)

Table 7: Part III against Parts I and II. Part III treats the oriented case $\theta = 1$ only.

its auxiliary recurrence, the entire Borel transform, and a simplified parameter-dependent index reconstruction. Two of these overlap Part II as special cases, and the reconstruction is an instance of the inversion apparatus Part I already cites (Remark 9.1); table 9 says where.

23.3 Notation

Throughout Part III the orientation weight is $\theta = 1$, and we drop the superscript: $X = X_{r,s,n}^{(1)}$, $G_n(u) = G_{r,s,n}^{(1)}(u) = \mathbb{E}(1+u)^X$, $A(n) = A_{r,s}^{(1)}(n)$, $B_J = B_J^{(1)}(u; r, s)$, $c_J = c_J^{(1)}(r, s)$. The moments $\mu_k = \mathbb{E}\binom{X}{k}$ are Part I's. The stable rational moments $\mu_k^*(n) = \mu_k^*(n; r, s)$ are the right side of (3.12) at $\theta = 1$, read as rational functions of an indeterminate n ; for $r = s = 2$ they are the μ_k^* of section 11. Likewise $R_h^{r,s}(n)$ is defined by (11.2) from $\mu_k^*(n; r, s)$, and $R_h^{2,2} = R_h$. Manuscript 03 overloads several letters that Parts I and II already use; table 8 fixes the convention. No normalization is changed.

Notion	Part III	Manuscript 03
count, statistic, generating function	$A(n), X, G_n(u)$	$a_{r,s}(n), X_{r,s,n}, G_n(u)$
offsets less one	$r-1, s-1$ (written out)	p, q (Part II's p_n is a polynomial)
sum of the offsets	$\sigma = r + s$	D (Part I's $\mathcal{D}$, D_v , D ; Part II's $D_{n,k}$)
weights $(r-1)^{\bar{j}}(s-1)^{\bar{j}}/j!$	$\mathbf{w}_j$	w_j (Part I's w is $\log z$)
profile; component of ℓ vertices	$\mathbf{b} = (b_\ell)_{\ell \geq 2}$	$(b_d)_{d \geq 1}$, $d = \ell - 1$ edges (Part I's d is the defect)
reciprocal-factorial functional	$\mathcal{L}_n^{(a)}, \bar{n} = n - k$	$\mathcal{L}_m^{(p)}$, $m = n - k$ (Part I's $\mathcal{L}$ is a law, m a dimer count)
profile series	$\mathcal{T}(z; y_1, y_2) = \tau_0 + \tau_1 y_1$	$T(z; y, w) = A + By$
auxiliary variables	$\zeta = z/(1-z); t, \xi$	ω (Part II's ω_j); t, v
moment cutoff; constants	$h_n = \lfloor \log n \rfloor; \gamma_{r,s}, \gamma_A$	$K; C_{p,q}, C_{r,s}, K_A$
inverse-factorial polynomials	$\Phi_i(u), \Phi(z, u)$	$P_m(u), \mathcal{P}(z, u)$ (Part I's $P_{\beta,j}$, forest $\mathcal{P}$)
their values at $u = -1$	$\psi_i, \Psi(z)$	$d_m, \mathcal{H}(z)$ (Part I's $d_j(w)$, $\mathcal{H}(t, D)$)
exponential generating function	$\mathcal{E}(\xi) = \sum \psi_i \xi^i / i!$	$\mathcal{E}(y)$
Borel transform of $C(x)$	$\widehat{C}(t)$	$\widehat{C}(t)$ (same)
inverse coordinates	$\Lambda, N, \Delta = \log N$ (Part II, $\theta = 1$)	T, z, ℓ
logarithmic coefficients	a_1, a_2	α, β
inverse corrections	ω_1, ω_2 (Part II)	A, B

Table 8: Notation of manuscript 03 against Part III. Other symbols ($R_h, \mu_k^*, B_J, c_J, F_r(n; \mathbf{b}), E_a(\mathbf{b}), \mathbf{b}!, E_M(n), S_j$) mean the same in both. In Part III, z is a formal variable, not Part I's Lambert backbone, which Part III calls N as Part II does.

Two of these renamings carry a tempting false reading. The weight $\mathbf{w}_j$ is not the logarithm $w = \log z$ of section 9, and $\psi_i = \Phi_i(-1)$ are not Part I's inverse coefficients $d_j(w)$: they are the coefficients of an inverse-factorial re-expansion of the avoidance probability (section 29).

24 A single sum for the stable moments

Put

$$\sigma = r + s, \quad \mathbf{w}_j = \frac{(r-1)^{\bar{j}}(s-1)^{\bar{j}}}{j!} \quad (j \geq 0). \quad (24.1)$$

Parts I and II	Manuscript 03's route
theorem 3.1, Corollary 3.2	the same proof (the identity (3.6) and the expansion of $\mathcal{D}^{-(r-1)}$ over subprofiles), credited to Part I and to Spahn–Zeilberger, with components indexed by their edge counts
(3.14)	the case $k = 1$ of the single-sum formula (remark 24.2)
Lemma 4.1, $\theta = 1$	the same argument; adds a bound on the normalized moments (lemma 27.1)
theorem 1.1, $\theta = 1$	a second proof through the collapse (theorem 27.2); its shape clause is Part I's, weaker than Part II's corollary 16.7
Corollary 7.3, $\theta = 1$	the same Cauchy argument on $ t = R$
(1.10), (7.3); proposition 17.2, $\theta = 1$	B_1, B_2, c_1, c_2 and c_3 for general (r, s) from the Stirling formula (remark 28.2)
proposition 18.1, corollary 18.2	the cases $r = 1$ or $s = 1$ of theorems 24.1, 25.1 and 28.1 (remark 25.2)
(8.7), section 8, table 1	the polynomials $B_0, \dots, B_8$ and the coefficients through order 16 at $(2, 2)$, equal to Part I's (section 31)
theorem 9.2, proposition 20.1, $\theta = 1$	the first two corrections in Part II's coordinates (proposition 32.1)
table 2	the same Bonferroni method at new sizes (table 11)

Table 9: Results of Parts I and II proved again by manuscript 03, and where Part III records each route.

Generalized binomial coefficients mean $\binom{a}{b} = a^b/b!$ for $b \geq 0$, also for negative integers a , and $\binom{a}{b} = 0$ for $b < 0$. The weights are nonnegative integers: if $r, s \geq 2$, then

$$w_j = j! \binom{r+j-2}{j} \binom{s+j-2}{j}, \quad (24.2)$$

and if $r = 1$ or $s = 1$, then $w_0 = 1$ and $w_j = 0$ for $j > 0$. For $r = s = 2$, $w_j = j!$.

Theorem 24.1 (Single-sum stable moments; manuscript 03). *For all $r, s \geq 1$ and $k \geq 0$, as an identity in $\mathbb{Q}(n)$,*

$$\mu_k^*(n) = \sum_{j=0}^k \frac{w_j \binom{n-\sigma+1-j}{k-j}}{n^{k+j}}. \quad (24.3)$$

Equivalently, for integers $n \geq 2k$,

$$\mu_k^*(n) = \frac{1}{n!} \sum_{j=0}^k w_j (n-k-j)! \binom{n-\sigma+1-j}{k-j}. \quad (24.4)$$

Equality with the actual moments μ_k additionally requires the stability conditions of Corollary 3.2.

The profile formula (3.12) has one term for each partition of k ; (24.3) has $k+1$.

24.1 A reciprocal-factorial linear functional

Put $\bar{n} = n - k$ and take n so large that $\bar{n} \geq k$. For an integer $a \geq 0$ and polynomials of degree at most $\bar{n}$ in a variable y , define

$$\mathfrak{L}_{\bar{n}}^{(a)}(y^i) = \frac{(-1)^i a^{\bar{i}}}{\bar{n}^i} \quad (0 \leq i \leq \bar{n}). \quad (24.5)$$

For $a = 0$ this is evaluation at $y = 0$. Since $(n - k - i)^{\overline{c-i}} = \bar{n}^c / \bar{n}^i$, the stable polynomial (3.2) becomes

$$F_r(n; \mathbf{b}) = \bar{n}^c \mathfrak{L}_{\bar{n}}^{(r-1)} \prod_{\ell \geq 2} (1 + (\ell - 1)y)^{b_\ell}. \quad (24.6)$$

Introduce the profile series

$$\begin{aligned} \mathcal{T}(z; y_1, y_2) &= 1 + \sum_{\ell \geq 2} z^{\ell-1} (1 + (\ell - 1)y_1)(1 + (\ell - 1)y_2) \\ &= \frac{1}{1 - z} + \frac{z(y_1 + y_2)}{(1 - z)^2} + \frac{z(1 + z)y_1 y_2}{(1 - z)^3}. \end{aligned} \quad (24.7)$$

In the multinomial expansion of $\mathcal{T}^{\bar{n}}$, a profile $\mathbf{b}$ of edge degree k has coefficient $\bar{n}^c / \mathbf{b}!$, and $(\bar{n}^c)^2 / n^{k+c} = \bar{n}! \bar{n}^c / n!$. With (24.6) for both forests, (3.12) at $\theta = 1$ becomes

$$\mu_k^*(n) = \frac{\bar{n}!}{n!} [z^k] \mathfrak{L}_{\bar{n}, y_1}^{(r-1)} \mathfrak{L}_{\bar{n}, y_2}^{(s-1)} \mathcal{T}(z; y_1, y_2)^{\bar{n}}. \quad (24.8)$$

24.2 The first functional

Write $\mathcal{T} = \tau_0 + \tau_1 y_1$. Since τ_1 has the factor z , the finite binomial theorem and (24.5) give, through degree $\bar{n}$ in z ,

$$\mathfrak{L}_{\bar{n}}^{(r-1)} (\tau_0 + \tau_1 y_1)^{\bar{n}} \equiv \tau_0^{\bar{n}} \sum_{i=0}^{\bar{n}} \frac{(-1)^i (r-1)^{\bar{i}}}{i!} \left(\frac{\tau_1}{\tau_0} \right)^i \equiv \frac{\tau_0^{\bar{n}+r-1}}{(\tau_0 + \tau_1)^{r-1}}. \quad (24.9)$$

The omitted terms start at $z^{\bar{n}+1}$, so they cannot affect $[z^k]$. With $\zeta = z/(1 - z)$ one has $\tau_0 = (1 + \zeta y_2)/(1 - z)$ and $\tau_0 + \tau_1 = (1 + 2\zeta y_2)/(1 - z)^2$, so the last expression is

$$(1 - z)^{r-1-\bar{n}} \frac{(1 + \zeta y_2)^{\bar{n}+r-1}}{(1 + 2\zeta y_2)^{r-1}}. \quad (24.10)$$

All its z -coefficients through degree $\bar{n}$ are polynomials in y_2 of degree at most $\bar{n}$, so the second functional is defined on them.

24.3 The second functional and the endpoint

For $s \geq 2$, a beta integral represents the second functional exactly:

$$\mathfrak{L}_{\bar{n}}^{(s-1)} f(y) = \frac{(\bar{n} + s - 1)!}{\bar{n}! (s - 2)!} \int_0^1 t^{s-2} (1 - t)^{\bar{n}} f\left(-\frac{t}{1 - t}\right) dt; \quad (24.11)$$

on y^i this follows from $\int_0^1 t^{s-2+i} (1 - t)^{\bar{n}-i} dt = (s - 2 + i)! (\bar{n} - i)! / (\bar{n} + s - 1)!$.

Assume first $r, s \geq 2$. Substitute (24.10) into (24.11) coefficientwise and then put $t = (1 - z)\xi$. For real $z < 0$ close to zero the result is

$$\frac{(\bar{n} + s - 1)!}{\bar{n}! (s - 2)!} (1 - z)^{\sigma-2-\bar{n}} \int_0^{1/(1-z)} \frac{\xi^{s-2} (1 - \xi)^{\bar{n}+r-1}}{[1 - (1 + z)\xi]^{r-1}} d\xi. \quad (24.12)$$

Only its Taylor coefficients through degree $\bar{n}$ are needed, not an equality of convergent series, and replacing the upper limit by 1 does not change them. To see the size of the change, put $a = -z > 0$: on the added interval $1 - \xi = O(a)$ and $1 - (1 - a)\xi \geq a/(1 + a)$, so the integral over it is $O(a^{\bar{n}+1})$. There is also a differentiation argument, not only a size estimate. For the integral with upper limit 1, differentiating $\ell' \leq \bar{n}$ times in z leaves an integrand bounded near $\xi = 1$ by a constant times $(1 - \xi)^{\bar{n}-\ell'}$, so dominated convergence computes its Taylor coefficients through degree $\bar{n}$. The integral before the substitution has the same property: with $a = -z$ and

$v = 1 - t$, its nonregular factor is $(v + a)^{\bar{n}+r-1}/(v + a(1+t))^{r-1}$, whose derivatives of order at most $\bar{n}$ are bounded by constant multiples of $(v + a)^{\bar{n}-\ell'}$. Its Taylor coefficients are therefore exactly the coefficientwise values of (24.11), and the change of endpoint is legitimate at every degree used.

Expand the denominator of the integrand at each fixed $0 \leq \xi < 1$:

$$\frac{(1 - \xi)^{\bar{n}+r-1}}{[1 - (1 + z)\xi]^{r-1}} = \sum_{j \geq 0} \frac{(r-1)^{\bar{j}}}{j!} z^j \xi^j (1 - \xi)^{\bar{n}-j}. \quad (24.13)$$

Only the terms with $j \leq \bar{n}$ are integrated, by the differentiation argument; they give $B(s-1+j, \bar{n}-j+1) = (s-2+j)! (\bar{n}-j)! / (\bar{n}+s-1)!$. Hence, as an equality of Taylor coefficients through degree $\bar{n}$,

$$\mathfrak{L}_{\bar{n}, y_1}^{(r-1)} \mathfrak{L}_{\bar{n}, y_2}^{(s-1)} \mathcal{T}^{\bar{n}} \equiv \frac{(1-z)^{\sigma-2-\bar{n}}}{\bar{n}!} \sum_{j=0}^{\bar{n}} w_j (\bar{n}-j)! z^j. \quad (24.14)$$

Nothing is claimed about coefficients beyond that degree.

If $r = 1$, evaluate at $y_1 = 0$ first. The remaining functional acts on $(1-z)^{-\bar{n}}(1+\zeta y_2)^{\bar{n}}$, and the same finite binomial calculation gives $(1-z)^{s-1-\bar{n}}$ through degree $\bar{n}$, which is (24.14) with $w_j = 0$ for $j > 0$. The case $s = 1$ follows by symmetry.

24.4 Completion of the proof

Proof of theorem 24.1 (manuscript 03). Insert (24.14) into (24.8) and use $\bar{n} = n - k$. The coefficient of z^{k-j} in $(1-z)^{\sigma-2-\bar{n}}$ is $\binom{\bar{n}-\sigma+2+k-j-1}{k-j} = \binom{n-\sigma+1-j}{k-j}$. This proves (24.4) for all large integers n . Both sides are rational functions of n , so the identity holds in $\mathbb{Q}(n)$ and gives (24.3). $\square$

Remark 24.2 (A first check; manuscript 03). For $k = 1$,

$$\mu_1^*(n) = \frac{n - \sigma + 1}{n} + \frac{(r-1)(s-1)}{n(n-1)} = \frac{(n-r)(n-s)}{n(n-1)},$$

which is Part I's exact mean (3.14) at $\theta = 1$. It confirms both the direction convention and the normalization.

25 Universal rational collapse

Theorem 25.1 (Universal rational-collapse identity; manuscript 03). *For all $r, s \geq 1$ and $h \geq 0$,*

$$R_h^{r,s}(n) = \sum_{j=0}^h \frac{w_j \binom{h-\sigma}{h-j}}{n^{\underline{h+j}}}. \quad (25.1)$$

In particular $R_0^{r,s} = 1$. For $h \geq \sigma$ the formula reduces to

$$R_h^{r,s}(n) = \sum_{j=\sigma}^h \frac{w_j \binom{h-\sigma}{j-\sigma}}{n^{\underline{h+j}}}, \quad (25.2)$$

and, when $r, s \geq 2$, its first term at infinity is

$$R_h^{r,s}(n) = w_\sigma n^{-h-\sigma} + O(n^{-h-\sigma-1}) \quad (h \geq \sigma). \quad (25.3)$$

When $r = 1$ or $s = 1$ it terminates:

$$R_h^{r,s}(n) = \begin{cases} (-1)^h \binom{\sigma-1}{h} / n^{\underline{h}}, & 0 \leq h \leq \sigma-1, \\ 0, & h \geq \sigma. \end{cases} \quad (25.4)$$

Proof (manuscript 03). Substitute (24.4) into (11.2), put $k = j + i$ and $H = h - j$, and interchange the two finite sums. For fixed j write $N' = n - 2j$ and $A' = n - \sigma + 1 - j$. The inner sum, before division by $n!$, is

$$\sum_{i=0}^H \frac{(-1)^{H-i}}{(H-i)!} (N' - i)! \binom{A'}{i} = (N' - H)! \sum_{i=0}^H (-1)^{H-i} \binom{A'}{i} \binom{N' - i}{H - i}. \quad (25.5)$$

The alternating Vandermonde identity

$$\sum_{i=0}^H (-1)^{H-i} \binom{A'}{i} \binom{N' - i}{H - i} = \binom{A' - N' + H - 1}{H} \quad (25.6)$$

holds because its left side is the coefficient of t^H in $\sum_i \binom{A'}{i} t^i (1-t)^{N'-i} = (1-t)^{N'} (1+t/(1-t))^{A'} = (1-t)^{N'-A'}$; generalized binomial series justify it coefficientwise. Here $A' - N' + H - 1 = h - \sigma$ and $N' - H = n - h - j$, so (25.5) equals $(n - h - j)! \binom{h - \sigma}{h - j}$. This proves (25.1) for large n , and then as an identity of rational functions.

For $h \geq \sigma$ the binomial coefficient has a nonnegative upper argument, so it vanishes unless $j \geq \sigma$; the $j = \sigma$ term has coefficient w_σ and denominator $n^{h+\sigma}$, and later terms are of higher order at infinity. If $r = 1$ or $s = 1$, only $j = 0$ survives, and $\binom{h-\sigma}{h} = (-1)^h \binom{\sigma-1}{h}$, which is zero for $h \geq \sigma$. $\square$

What the cancellation explains. Before normalization each fixed moment has the nonzero Poisson limit $1/k!$. After multiplication by e^{-u} , low powers of $1/n$ still occur for $h < \sigma$, but every normalized coefficient with $h \geq \sigma$ loses the first σ possible powers of $1/n$. The loss is exact and uniform in h : it is encoded by the vanishing binomial coefficients with $j < \sigma$. This explains the late appearance of high powers of u in the correction polynomials. The formula is symmetric in r, s , which also follows combinatorially: $\pi \mapsto \pi^{-1}$ exchanges a directed position gap r and a directed value gap s (table 5 records the same observation from Part II's source).

Remark 25.2 (The one-path case is Part II's Proposition 18.1). [Note of the batch-85 write.] For $r = 1$, (24.3) reduces to $\mu_k^*(n) = \binom{n-s}{k}/n^k$, which is (18.1), and (25.4) gives the finite inverse-factorial sector

$$e^{-u} G_n(u) \sim \sum_{h=0}^s \frac{(-1)^h \binom{s}{h} u^h}{n^h}, \quad (25.7)$$

which is the rational factor of (18.2) (for $s = 2$, $1 - 2u/n + u^2/(n(n-1))$). Manuscript 03 derives both as special cases of its general theorems and credits the one-path inclusion–exclusion to Spahn and Zeilberger [3], but does not cite Part II's proposition, which proves the same statements directly; the credit belongs to Part II (manuscript 03 of batch 74). At $u = -1$ the sector is $\sum_{h=0}^s \binom{s}{h}/n^h$; for $r = s = 1$ it is $1 + 1/n$, the values of Remark 7.2. For $(r, s) = (1, 3)$ the ordinary inverse-power expansion is infinite, $c_J = 1, 3, 3, 4, 6, 10, 18, 34, \dots$, although the inverse-factorial representation has four terms.

26 The A189281 conjecture is a special case

For $r = s = 2$, $\sigma = 4$ and $w_j = j!$. At $h \leq 3$, (25.1) gives the four rational functions $R_0, \dots, R_3$ displayed before Conjecture 11.1, and every later order collapses further.

Theorem 26.1 (Resolution of the rational-collapse conjecture; manuscript 03). *For every integer $h \geq 4$,*

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^{\frac{5}{2}}}. \quad (26.1)$$

That is, Conjecture 11.1 holds, and so does the ratio recurrence (11.8).

Proof (manuscript 03). For large integers n , (25.2) gives $n!R_h(n) = \sum_{j=4}^h j! \binom{h-4}{j-4} (n-h-j)!$. Put $H = h - 4$, $N' = n - h - 4$ and $j = i + 4$, so that the sum is $\sum_{i=0}^H \binom{H}{i} (i+4)! (N' - i)!$. With $M' = N' - H \geq 0$ this is

$$24H!M'! \sum_{i=0}^H \binom{i+4}{4} \binom{M' + H - i}{M'},$$

and the convolution equals $\binom{M'+H+5}{H}$, the coefficient of x^H in $(1-x)^{-5}(1-x)^{-M'-1}$. Hence

$$n!R_h(n) = \frac{24(N' - H)!(N' + 5)!}{(N' - H + 5)!} = \frac{24(n - 2h)!(n - h + 1)!}{(n - 2h + 5)!}.$$

Dividing by $n!$ gives (26.1); rational continuation removes the restriction to large n , and dividing consecutive instances gives (11.8). $\square$

This proves the conjecture for every h ; it does not merely extend the finite range $h = 4, \dots, 16$ checked in Part I. It also explains the constant: the Vandermonde convolution starts with a component-index shift of four, which produces $4! = 24$. Part I's question 1 asked for both a proof and an explanation of the factor 24 and of the loss of four powers of n ; the loss is the case $\sigma = 4$ of the cancellation above. It asked for a telescoping or combinatorial proof; the proof given is a single-sum identity followed by Vandermonde convolutions, and a bijective explanation is still open (section 34, question 9).

Remark 26.2 (A recurrence that is not proved here; manuscript 03). Equation (11.8) is a recurrence in the moment order h for rational functions of n . It is not the OEIS recurrence in the permutation size n for A189281(n). Confusing the two would turn a correct identity into an unsupported claim about the sequence. The order-eight, degree-eleven recurrence of A189281 remains unproved.

Corollary 26.3 (Proposition 11.2 is unconditional). $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$ for every J , every $c_J^{(1)}(2, 2)$ is an integer, and $\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J - 4)$ for $J \geq 4$.

The write's statement. The proof of Proposition 11.2 used only Conjecture 11.1, which theorem 26.1 proves. Manuscript 03 proves more, by a second route: theorem 28.1 at $(r, s) = (2, 2)$, which also shows that the degree bound is attained for $J \geq 8$. $\square$

27 From the collapse to the asymptotic expansion

Manuscript 03 proves the all-orders theorem again, directly from the collapse and without assuming Part I's theorem. The route is printed here because it is genuinely different from Part I's: instead of the defect filtration and finite differences in the number of dimers, it bounds the normalized moments R_h uniformly and expands finitely many rational functions.

Lemma 27.1 (Moment bounds; manuscript 03). *For arbitrary directed position and value path forests on n vertices, $0 \leq \mu_k \leq e^{2k}/k!$, and the actual normalized moments $\tilde{R}_h(n) = [u^h]e^{-u}G_n(u)$ satisfy*

$$|\tilde{R}_h(n)| \leq \frac{(1 + e^2)^h}{h!}. \quad (27.1)$$

Proof (manuscript 03). The first bound is Lemma 4.1 at $\theta = 1$, with the same proof. Then $|\tilde{R}_h| \leq \sum_{k=0}^h e^{2k}/(k!(h-k)!) = (1 + e^2)^h/h!$. $\square$

Theorem 27.2 (Uniform expansion through the collapse; manuscript 03). *For fixed $r, s \geq 1$, every $M \geq 0$ and every finite U , uniformly on $|u| \leq U$,*

$$G_n(u) = e^u \left(\sum_{J=0}^M \frac{B_J(u)}{n^J} + O_{M,r,s,U}(n^{-M-1}) \right), \quad (27.2)$$

where

$$B_0 = 1, \quad B_J(u) = \sum_{h \geq 1} [x^J] R_h^{r,s}(1/x) u^h \quad (J \geq 1), \quad (27.3)$$

each sum being finite. These are Part I's $B_J^{(1)}(u; r, s)$, by uniqueness of asymptotic coefficients; in particular $B_J(0) = 0$ for $J > 0$. The same polynomials apply to forests whose shortest path is at least a fixed positive fraction of n .

Second proof of [theorem 1.1](#) at $\theta = 1$ (manuscript 03). Let $h_n = \lfloor \log n \rfloor$. For large n the stable range contains every $h \leq h_n$, so by (27.1), with $\tilde{a} = (1 + e^2)U$,

$$e^{-u} G_n(u) = \sum_{h=0}^{h_n} R_h^{r,s}(n) u^h + O\left(e^{\tilde{a}} \frac{\tilde{a}^{h_n+1}}{(h_n+1)!}\right), \quad (27.4)$$

and the error is smaller than every power of $1/n$, its logarithm being $-\log n \log \log n + O(\log n)$.

Suppose $r, s \geq 2$. For $\sigma \leq h \leq h_n$ and $n > 2h_n$ all terms of (25.2) are nonnegative. Uniformly in $j \leq h \leq h_n$, $1/n^{h+j} \leq 2n^{-h-j}$ for large n , because the product of the factors $1 - i/n$ with $i < 2h_n$ is $1 + O(h_n^2/n)$. Moreover $w_{j+1}/w_j = (r-1+j)(s-1+j)/(j+1) \leq \gamma_{r,s}(h_n+1)$ for $\sigma \leq j < h_n$, so that $\sum_{j=\sigma}^{h_n} w_j n^{-j} \leq 2w_\sigma n^{-\sigma}$ for large n . Bounding the binomial coefficients by 2^h gives

$$|R_h^{r,s}(n)| \leq \gamma_{r,s} 2^h n^{-h-\sigma} \quad (\sigma \leq h \leq h_n). \quad (27.5)$$

Put $h_* = \max(\sigma - 1, M - \sigma)$. Summing (27.5) against $|u|^h$ over $h_* < h \leq h_n$ gives $O(n^{-M-1})$. Only the finite family $R_0, \dots, R_{h_*}$ remains; each is rational in n , regular at $n = \infty$, with an ordinary Taylor remainder there. Expanding them proves (27.2) and (27.3). If $r = 1$ or $s = 1$, (25.4) makes the sum in (27.4) terminate at $h = \sigma - 1$, and the conclusion is immediate. For other sufficiently long forests the stability theorem and the moment bound are unchanged. Uniqueness follows by multiplying the difference of two expansions by successive powers of n and taking limits. $\square$

The identity (27.3) is the one Part I's proof of Proposition 11.2 uses at (2, 2). The shape clause of [theorem 27.2](#) is the one stated after Part I's proof of [theorem 1.1](#); it is weaker than Part II's [theorem 16.6](#) and [corollary 16.7](#) and [Remark 16.8](#), which need only $L_{\min}(n) \log \log n / \log n \rightarrow \infty$, and it is not an improvement on them. Evaluating at $u = -1$ gives (1.11) with $c_J = B_J(-1)$, and the Cauchy argument on $|t| = R > 1$ gives the whole-law statement of [Corollary 7.3](#) at $\theta = 1$; manuscript 03 says that this mechanism was already in Part I and that what the collapse adds is the arithmetic structure of the polynomials.

28 Integer coefficients and an exact degree law

The inverse falling factorial has the expansion

$$\frac{1}{(1/x)^{\overline{m}}} = \frac{x^m}{\prod_{i=1}^{m-1} (1 - ix)} = \sum_{J \geq m} \left\{ \begin{matrix} J-1 \\ m-1 \end{matrix} \right\} x^J \quad (m \geq 1), \quad (28.1)$$

where $\left\{ \begin{smallmatrix} a \\ b \end{smallmatrix} \right\}$ is the Stirling number of the second kind, zero outside $0 \leq b \leq a$, with $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$.

It follows by induction from $\left\{ \begin{smallmatrix} a \\ b \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} a-1 \\ b-1 \end{smallmatrix} \right\} + b \left\{ \begin{smallmatrix} a-1 \\ b \end{smallmatrix} \right\}$, or by multiplying the generating functions of the columns.

Theorem 28.1 (Universal integrality and exact degrees; manuscript 03). *For all $r, s \geq 1$ and $J \geq 1$,*

$$B_J(u) = \sum_{h=1}^J u^h \sum_{j=0}^h w_j \binom{h-\sigma}{h-j} \left\{ \begin{smallmatrix} J-1 \\ h+j-1 \end{smallmatrix} \right\}. \quad (28.2)$$

In particular,

$$B_J^{(1)}(u; r, s) \in \mathbb{Z}[u], \quad c_J^{(1)}(r, s) \in \mathbb{Z} \quad \text{for every } J \geq 0. \quad (28.3)$$

The degree satisfies

$$\deg_u B_J \leq \min\{J, \max(\sigma-1, J-\sigma)\}, \quad (28.4)$$

and if $r, s \geq 2$ and $J \geq 2\sigma$ it is exact:

$$\deg_u B_J = J - \sigma, \quad [u^{J-\sigma}]B_J = w_\sigma. \quad (28.5)$$

All coefficients of u^h with $h \geq \sigma$ are nonnegative. When $r = 1$ or $s = 1$, $\deg B_J \leq \sigma - 1$ for all J .

Proof (manuscript 03). Substitute (25.1) and (28.1) into (27.3); this is (28.2). Every factor of every summand is an integer, the generalized binomial coefficient with a possibly negative upper argument included, so integrality follows without a denominator-clearing bound. For $h < \sigma$, $R_h(1/x)$ begins no earlier than x^h ; for $h \geq \sigma$, (25.2) begins no earlier than $x^{h+\sigma}$. This proves (28.4). If $J \geq 2\sigma$, the largest possible h is $J - \sigma \geq \sigma$, where only $j = \sigma$ contributes, with coefficient $w_\sigma \binom{J-2\sigma}{J-2\sigma} \left\{ \begin{smallmatrix} J-1 \\ J-1 \end{smallmatrix} \right\} = w_\sigma > 0$. For $h \geq \sigma$ every summand of (28.2) is nonnegative. The one-offset assertion follows from (25.4). $\square$

Only the summation over $j \leq \min(h, J-h)$ contributes, because $\left\{ \begin{smallmatrix} J-1 \\ h+j-1 \end{smallmatrix} \right\} = 0$ for $h+j > J$. For A189281, (28.5) says that for every $J \geq 8$

$$\deg B_J^{(1)}(u; 2, 2) = J - 4, \quad [u^{J-4}]B_J^{(1)}(u; 2, 2) = 24, \quad (28.6)$$

so Part I's conditional bound $\max(3, J-4)$ is attained. Moreover every coefficient of u^h with $h \geq 4$ is divisible by 24, because only the weights $j!$ with $j \geq 4$ contribute. No sign rule for $c_J = B_J(-1)$ is inferred: evaluation at $u = -1$ reintroduces cancellation.

Two earlier questions are answered here. Part I's question 1 and Part II's overlapping question asked whether all $c_J^{(1)}(2, 2)$ are integers, and whether $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for all r, s : both hold. Part II's question 1 asked whether $\deg_u B_J^{(1)}(u; r, s) \leq J$ at every order: it holds, by (28.4). The absolute case $\theta = 2$, whose coefficients are not integers by (8.12), is not covered, and Proposition 7.1 remains its denominator statement.

28.1 The first corrections for arbitrary offsets

Formula (28.2) gives

$$\begin{aligned} B_1 &= -(\sigma - 1)u, & B_2 &= (r - 1)(s - 1)u + \binom{\sigma - 1}{2}u^2, \\ B_3 &= (r - 1)(s - 1)u + \left(\binom{\sigma - 1}{2} - (\sigma - 2)(r - 1)(s - 1) \right)u^2 - \binom{\sigma - 1}{3}u^3, \end{aligned} \quad (28.7)$$

and consequently

$$c_0 = 1, \quad c_1 = \sigma - 1, \quad c_2 = \binom{\sigma - 1}{2} - (r - 1)(s - 1), \quad c_3 = \binom{\sigma}{3} - (\sigma - 1)(r - 1)(s - 1). \quad (28.8)$$

For example $c_3 = 0$ at $(3, 3)$, but $c_3 = -7$ and -24 at $(4, 4)$ and $(5, 5)$; such cancellations are outputs of the formula, not extrapolations from data.

Remark 28.2 (These are Parts I and II's first corrections). [Note of the batch-85 write.] B_1 and B_2 are Part I's (1.10) and (7.3) at $\theta = 1$. The values c_1, c_2, c_3 are those of Part II's [proposition 17.2](#) at $\theta = 1$, which manuscript 03 does not cite: $\binom{\sigma - 1}{2} - (r - 1)(s - 1) = \frac{1}{2}(r^2 - r + s^2 - s)$, and

$$\binom{\sigma}{3} - (\sigma - 1)(r - 1)(s - 1) = \frac{1}{6}(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6),$$

which is (17.1); the intake checked the identity algebraically and the write checked both values for $1 \leq r, s \leq 5$ against (28.2). The closed form of $c_3^{(1)}(r, s)$ is therefore Part II's result (manuscript 03 of batch 74), here re-derived as a special case; the absolute $c_3^{(2)}$ of [proposition 17.2](#) has no counterpart in Part III.

29 An inverse-factorial kernel and an auxiliary recurrence

Define integer polynomials

$$\Phi_i(u) = \sum_{j=0}^{\lfloor i/2 \rfloor} w_j \binom{i - j - \sigma}{i - 2j} u^{i-j} \quad (i \geq 1), \quad \Phi_0(u) = 1. \quad (29.1)$$

Then the algebraic asymptotic sector of $e^{-u}G_n(u)$ is

$$\sum_{i \geq 0} \frac{\Phi_i(u)}{n^i}, \quad (29.2)$$

in the sense of equality after re-expansion in $1/n$, coefficient by coefficient. The displayed sum is not an ordinary convergent series at an integer n ; its denominators vanish for $i > n$.

Proposition 29.1 (Formal hypergeometric kernel; manuscript 03). *As a formal power series in z with polynomial coefficients in u ,*

$$\Phi(z, u) := \sum_{i \geq 0} \Phi_i(u) z^i = (1 - uz)^{\sigma-1} \sum_{j \geq 0} w_j \left(\frac{uz^2}{1 - uz} \right)^j = (1 - uz)^{\sigma-1} {}_2F_0\left(r - 1, s - 1; ; \frac{uz^2}{1 - uz}\right), \quad (29.3)$$

where ${}_2F_0(a, b; ; v) = \sum_{j \geq 0} a^{\bar{j}} b^{\bar{j}} v^j / j!$ is a formal series.

Proof (manuscript 03). In (25.1) write $h = j + i'$ and $i = h + j$. For fixed j the remaining sum is $\sum_{i' \geq 0} \binom{j + i' - \sigma}{i'} (uz)^{i'} = (1 - uz)^{\sigma-j-1}$. Multiply by $w_j u^j z^{2j}$ and sum over j . Each coefficient of z uses finitely many terms, so the rearrangements are formal and well defined. $\square$

Avoidance coefficients as a Stirling transform. Put $\psi_i = \Phi_i(-1)$ and

$$\Psi(z) = \sum_{i \geq 0} \psi_i z^i = (1+z)^{\sigma-1} \sum_{j \geq 0} w_j \left(-\frac{z^2}{1+z} \right)^j. \quad (29.4)$$

Then

$$c_0 = \psi_0 = 1, \quad c_J = \sum_{i=1}^J \psi_i \left\{ \begin{matrix} J-1 \\ i-1 \end{matrix} \right\} \quad (J \geq 1). \quad (29.5)$$

The inverse-power coefficients are thus a shifted Stirling transform of an integer sequence with a compact formal generating function, an alternative to constructing every B_J . For $r, s \geq 2$ the ${}_2F_0$ series has radius of convergence zero in its argument, because $w_{j+1}/w_j \sim j$. That does not invalidate (29.3), since formal substitution into a series of positive valuation is legitimate; it does forbid treating the formula as an analytic generating function without a stated summation procedure.

A short recurrence at (2, 2). For $r = s = 2$ write $\mathcal{F}(t) = \sum_{j \geq 0} j! t^j$ (a formal series in a local variable t). Comparing coefficients proves

$$t^2 \mathcal{F}'(t) + (t-1)\mathcal{F}(t) = -1. \quad (29.6)$$

Substituting $t = -z^2/(1+z)$ into $\Psi(z) = (1+z)^3 \mathcal{F}(t)$ and using the chain rule gives

$$z^3(1+z)\Psi'(z) + (2+3z+3z^2-2z^3)\Psi(z) = (z+2)(z+1)^4, \quad (29.7)$$

and hence

$$\boxed{2\psi_i + 3\psi_{i-1} + (i+1)\psi_{i-2} + (i-5)\psi_{i-3} = 0 \quad (i \geq 6),} \quad (29.8)$$

with $(\psi_0, \dots, \psi_5) = (1, 3, 2, -1, 1, 2)$; the next values are $-6, 0, 24, -24, -96, 216, 384, -1704, -1296, 13656$. This is an exact recurrence for the *auxiliary* inverse-factorial coefficients ψ_i , not for $A189281(n)$. It generates $\psi_0, \dots, \psi_M$ in $O(M)$ integer operations, and the Stirling transform then takes $O(M^2)$; these are counts of arithmetic operations, not bit-complexity bounds. For general fixed offsets, (29.1) gives $\psi_0, \dots, \psi_M$ in $O(M^2)$ operations, and (28.2) gives the whole array B_J in $O(M^3)$, with no enumeration of partitions and no recurrence guessing. [Added in the batch-85 write: the write recomputed Ψ from (29.4) to order 39 in exact arithmetic and found (29.8) satisfied for $6 \leq i \leq 39$, with the values above.]

30 An entire Borel transform of the algebraic sector

Define the exponential generating function

$$\mathcal{E}(\xi) = \sum_{i \geq 0} \frac{\psi_i}{i!} \xi^i. \quad (30.1)$$

Unlike Ψ , it is a genuine entire function. Recall Part I's formal series $C(x) = 1 + \sum_{j \geq 1} c_j x^j$ (section 9).

Theorem 30.1 (Entire Borel transform; manuscript 03). *For all fixed $r, s \geq 1$ the series (30.1) is entire, and the order-one Borel transform of the nonconstant part of C is the entire function*

$$\boxed{\hat{C}(t) := \sum_{J \geq 1} \frac{c_J}{(J-1)!} t^{J-1} = \mathcal{E}'(e^t - 1).} \quad (30.2)$$

In particular, for every $A > 0$ there is a constant γ_A such that

$$|c_J| \leq \gamma_A A^{J-1} (J-1)! \quad (J \geq 1). \quad (30.3)$$

Proof (manuscript 03). Suppose $r, s \geq 2$. By (24.2), $w_j \leq \gamma_{r,s}(j+1)^{\sigma-4}j!$. For large i and $j \leq i/2$, the binomial coefficient in (29.1) has a nonnegative upper argument and modulus at most 2^i ; the finitely many remaining i are absorbed in the constant. Hence

$$|\psi_i| \leq \gamma_{r,s} 2^i (i+1)^{\sigma-3} [i/2]!. \quad (30.4)$$

The i th root of $[i/2]!/i!$ tends to zero, so (30.1) converges for every complex ξ . If $r = 1$ or $s = 1$, $\Psi(z) = (1+z)^{\sigma-1}$ is a polynomial and the conclusion is immediate. The exponential generating function of the Stirling numbers, $\sum_{J \geq i} \left\{ \begin{smallmatrix} J-1 \\ i-1 \end{smallmatrix} \right\} t^{J-1}/(J-1)! = (e^t - 1)^{i-1}/(i-1)!$, combined with (29.5), gives (30.2). The interchange is justified locally by the absolute convergence of the entire series for $\mathcal{E}'$, and the composition is entire. Cauchy's estimate on $|t| = 1/A$ gives (30.3). $\square$

What this does not imply. An entire Borel transform need not grow slowly enough for a Laplace integral along the positive axis, and composition with $e^t - 1$ can create very rapid growth. Theorem 30.1 is therefore neither a proof of Borel–Laplace summability nor a proof that $\sum c_J n^{-J}$ converges. Nor can the algebraic sector determine contributions smaller than every power of $1/n$: different long path forests can share their stable moments to increasing order, hence the whole array B_J , while their exact counts differ (Remark 16.4). The theorem describes the algebraic sector and its Borel transform, not the beyond-all-orders corrections; this is the boundary between the results proved here and a full transseries theory. It bears on Part I's question 4, which asks for growth bounds in M and a justified truncation rule; those remain open (section 34, questions 3 and 4).

31 Explicit OEIS consequences

A189281. The polynomials $B_0, \dots, B_8$ that (28.2) gives at $(2, 2)$ are Part I's (section 8); the term u^4 first appears at order eight, in $B_8 = u + 479u^2 + 378u^3 + 24u^4$, with coefficient 24, as (28.6) requires. The avoidance expansion through n^{-10} is Part I's (8.7), with $c_4 = 0$; manuscript 03 says expressly that these displayed terms were already proved in Part I and that its new content is the all-orders identity and the integrality theorem, not these finite terms. Its coefficients through order 16 equal table 1. Beyond them, (28.2) gives

J	$c_J^{(1)}(2, 2)$	J	$c_J^{(1)}(2, 2)$
17	−9526718817	21	57992423942395
18	−136484458282	22	1360005443948778
19	−1110549722719	23	16301210950936317
20	−3629525434664	24	113926603325586836

and `data/03-rational-collapse-coefficients.json` holds every B_J , c_J and ψ_J through order 100. The runs of alternating signs neither undermine integrality nor indicate a proved periodic sign pattern; large-order sign and growth behaviour is a separate problem.

Three further diagonal sequences. The theorem applies directly to the OEIS entries for the directed gaps $(3, 3)$, $(4, 4)$ and $(5, 5)$, A189282, A189283 and A189284 [15, 16, 17], whose pages display expansions through n^{-2} . For every fixed truncation order the rows of table 10 continue by the same finite formula, with a rigorous next-order remainder.

For the nondiagonal pair $(r, s) = (2, 3)$ the coefficients through order eight are 1, 4, 4, 2, −1, 6, 81, 388, 775, so the result does not depend on equal offsets. The data file also holds examples with one offset equal to one, as degenerate checks (remark 25.2).

OEIS	r	s	c_1	c_2	c_3	c_4	c_5	c_6
A189281	2	2	3	2	1	0	3	26
A189282	3	3	5	6	0	-13	-3	235
A189283	4	4	7	12	-7	-73	-88	1029
A189284	5	5	9	20	-24	-234	-390	3374

Table 10: Manuscript 03: $A_{r,r}^{(1)}(n)/n! = e^{-1}(1 + \sum_{J \geq 1} c_J n^{-J})$ in the Poincaré sense; not a claim that the series converges. The first row is Part I's (8.7).

32 Index reconstruction with Lambert W

Manuscript 03 specializes the repository's inversion apparatus, which it cites through this report, to a simple parameter-dependent formula along exact sequence values, so that no interpolation of the integer sequence is chosen. In Part II's coordinates (20.1) at $\theta = 1$, for large y put

$$\Lambda = \log y - \kappa_1 = \log y + 1 - \frac{1}{2} \log(2\pi), \quad N = \frac{\Lambda}{W_0(\Lambda/e)}, \quad \Delta = \log N, \quad (32.1)$$

so that $N(\log N - 1) = \Lambda$.

Proposition 32.1 (Index reconstruction; manuscript 03). *For $y = A_{r,s}^{(1)}(n)$ and fixed offsets, as $n \rightarrow \infty$,*

$$n = N - \frac{1}{2} + \frac{1/24 - (\sigma - 1)}{N\Delta} + \frac{(r-1)(s-1)}{N^2\Delta} + O_{r,s}\left(\frac{1}{N^3\Delta}\right). \quad (32.2)$$

Every further fixed-order correction is computable from (28.2) and the Stirling expansion of $\log \Gamma$.

Proof (manuscript 03). The expansion (1.11) and Stirling's expansion [7] give, as in (9.3), $\Lambda = n(\log n - 1) + \frac{1}{2} \log n + a_1/n + a_2/n^2 + O(n^{-3})$ with $a_1 = c_1 + \frac{1}{12}$ and $a_2 = c_2 - \frac{1}{2}c_1^2$. Write $n = N + \delta$ and expand about N . The constant order gives $\delta = -\frac{1}{2}$; with $\delta = -\frac{1}{2} + \omega_1/N + \omega_2/N^2$ the next two coefficients are $\omega_1\Delta + a_1 - \frac{1}{8}$ and $\omega_2\Delta + a_2 + \frac{1}{2}a_1 - \frac{1}{24}$. They vanish for $\omega_1 = (1/24 - c_1)/\Delta$ and $\omega_2 = (c_1(c_1 - 1)/2 - c_2)/\Delta = (r-1)(s-1)/\Delta$, by (28.8). The residual is $O(N^{-3})$ and the smooth main part has derivative $\log N + O(1/N)$, so the mean value theorem turns it into an index error $O(N^{-3}/\log N)$. The initial bound $n - N = O(1)$ follows from the same equation and the monotonicity of $x(\log x - 1)$ for large x . $\square$

[Note of the batch-85 write.] This is Part II's [proposition 20.1](#) at $\theta = 1$, with the same coordinates and the same two coefficients ($1/8 - a_1 = 1/24 - c_1$, and $1/24 - a_1/2 - a_2 = (r-1)(s-1)$); what is new is that the second coefficient is written in closed form for every pair of offsets. At (2, 2) it is the $1/(N^2\Delta)$ of (20.4). In Part I's coordinates it is [theorem 9.2](#), and the method is the inversion apparatus of [Remark 9.1](#), an instance of `p0:thm:perturbed-inversion` of [12]; no novelty is claimed for the method.

For the exact A189281 values at $n = 20$ and $n = 30$ the displayed approximation misses n by about $2.05663 \cdot 10^{-4}$ and $5.16805 \cdot 10^{-5}$ (the first agrees with Part II's [section 21](#), which evaluates the same formula). These are numerical diagnostics, not certified rounding thresholds. The proposition implies eventual nearest-integer recovery along exact sequence values, but gives no explicit onset without an additional effective remainder estimate.

33 Manuscript 03's exact checks and certificates

Manuscript 03 separates proof, exact finite algebra and numerical diagnostics; its proofs of arbitrary order are those of [sections 24](#) and [25](#), and its checks exercise independent implementations to detect transcription or indexing errors. Its `verify.py` (shipped as `code/03-rational-collapse-verify.py`, standard library only) builds F_r directly from [\(3.2\)](#) and sums over integer partitions, and separately implements [\(25.1\)](#); each comparison clears the denominator by multiplying by $h!n^{2h}$ and compares integer polynomials in n . A separate finite tiling routine enumerates ordered tile-size multisets on each actual path, without the stable polynomial or the single-sum theorem, and a third routine enumerates permutations and records the whole violation histogram. The recorded run passed:

Exact check	Number
Polynomial identities between profile sums and (25.1) , $h = 0, \dots, 16$, nine offset pairs	153
Complete violation histograms against brute permutations, $n = 0, \dots, 8$, $r, s \in \{1, 2, 3\}$	81
Single-sum moments against brute histograms in the stable range	190
A189281 values, $n = 0, \dots, 30$, from independent finite tilings	31
The recurrence (29.8) , $i = 6, \dots, 100$	95
(29.5) against the polynomials B_J , $J = 1, \dots, 100$	100
Sharp-degree and leading-coefficient checks across offset pairs	178

The nine offset pairs were $(1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (4, 4)$. It also compared [\(26.1\)](#) with the general expression for $h = 4, \dots, 16$, and the first seventeen $c_J(2, 2)$ with Part I's values. The A189281 values come from a supplied excerpt of the OEIS b-file [\[18\]](#); they are validation targets only, never inputs to coefficient generation or to any proof. The program computes all B_J , c_J and ψ_J at $(2, 2)$ through order 100 in exact integers. [Added in the batch-85 write: if the b-file excerpt is missing under its delivered name, the program skips those 31 checks without failing and prints no line for them; the report README gives a rerun recipe that restores the delivered names.]

Rational Bonferroni certificates. The single-sum formula gives exact low moments without computing the full count. With the inequalities [\(10.1\)](#) and the scaled residual [\(10.2\)](#) at $\theta = 1$, manuscript 03's `certify.py` uses S_{31} and S_{30} at $n = 64, 128, 256$, where these moments are in the stable range, and brackets e^{-1} by its alternating Taylor sums through degrees 101 and 102. Dividing the probability bounds by that positive rational interval gives exact rational intervals, rounded outward in [table 11](#); the unrounded fractions are in `data/03-rational-collapse-bonferroni.json`.

n	Certified interval for $E_4(n)$	Certified interval for $E_8(n)$
64	[3.43128403, 3.43128404]	[−1630.56920464, −1630.56920463]
128	[3.20934303, 3.20934304]	[−1516.24227317, −1516.24227316]
256	[3.10311068, 3.10311069]	[−1461.72210309, −1461.72210308]

Table 11: Manuscript 03: exact finite certificates for A189281 ($\theta = 1$), generated without a recurrence for A189281(n). The limits 3 and -1409 follow from the asymptotic theorem, not from extrapolating three rows. They interleave with Part I's intervals at $n = 80, 160, 320$ ([table 2](#)).

Manuscript 03's optional inverse diagnostic, `inverse_demo.py`, needs `mpmath`; neither the verifier nor the mathematical results depend on it. Repeating the finite tests to a larger order

adds implementation checks, not a substitute for the proofs, and no Lean kernel run is asserted. Its delivered rerun commands use the delivered layout; the report README gives the commands for the shipped names.

34 Further questions raised by manuscript 03

Manuscript 03 poses ten research projects, as proposals only, ordered to separate exact finite enumeration, the algebraic sector and beyond-all-orders information. Several continue questions of Parts I and II, as noted.

1. *Prove the original A189281 recurrence* (Part I's question 2). The order-eight, degree-eleven recurrence remains a separate target. One route writes the exact finite tiling sum as its stable contribution plus explicit boundary corrections and seeks a telescoping certificate for the whole; agreement of its formal asymptotic solution with (26.1) is a consistency check, not a proof. The first deliverable should be a finite identity whose boundary terms are fully specified.
2. *The first beyond-all-orders term* (Part I's question 3, Part II's question 2). Find the leading difference between two forest shapes with the same r, s, n and long paths, whose low moments agree until a component can meet a whole short path: is it a single spanning component or a family of competing profiles? A first case changes two path lengths by $+1$ and -1 .
3. *Large-order growth of c_J* (Part I's question 4). Use (30.3) and the composition $\mathcal{E}'(e^t - 1)$ to find sharp asymptotics, oscillation and signs, first at $(2, 2)$ and then for all fixed offsets; (29.8) may be a first step.
4. *Summation and optimal truncation* (Part I's question 4). Determine the growth of $\hat{C}$ in sectors and whether an accelerated or nonstandard Laplace transform recovers a natural analytic object; a complete answer names the function reconstructed, the directions, the optimal truncation order in n , and the relation to the finite counts.
5. *A finite-algebraic proof of the single-sum lemma*. The beta integral is used only to evaluate a finite Taylor jet; a telescoping proof of (24.14), a polynomial identity for the reciprocal-factorial functionals, would simplify formalization and may show a direct combinatorial cancellation.
6. *Formalize the finite core* (Part I's question 7, Part II's formal package). A Lean development can first define falling products, generalized integer binomials and (24.3), (25.1); after clearing denominators the collapse is a finite rearrangement and a Vandermonde convolution. Stable tilings and the analytic remainder are further layers, and a formal proof of the algebraic identity alone must not be labelled a formal proof of the asymptotic statement.
7. *Absolute differences*. For $|\pi(i+r) - \pi(i)| \neq s$ each component has two orientations (weight 2^c , not 2^k), and the cancellation behind (25.1) changes. Find an equally explicit kernel, optimal denominator bounds, and which arithmetic survives; the directed integrality theorem must not be transplanted without proof.
8. *Growing offsets* (Part I's question 5). If σ grows, the sharp-degree threshold, the weight w_σ and the stability range move. Find ranges such as $r + s = o(\log n)$ with a uniform theorem, and the crossovers where Poisson(1) normalization fails; the exact mean $(n - r)(n - s)/(n(n - 1))$ constrains any scaling limit.

9. *Positivity and divisibility, combinatorially.* For $h \geq \sigma$ the coefficients of u^h in B_J are nonnegative sums of integer weights and Stirling numbers; find a labelled-object interpretation. At $(2, 2)$ the factor 24 suggests a four-element ordering or symmetry, but the proof gives a convolution identity, not a bijection (compare Part I's question 1).
10. *Several forbidden gaps* (Part I's question 6, Part II's question 4). The graphs are then not unions of paths, and components may branch or cycle. Determine which sparse families still admit stable component polynomials and a finite hypergeometric normalization; a first project is a two-gap directed model of bounded component width.

Manuscript 03 suggests starting with questions 5 and 6 and the asymptotics of (29.8); questions 1 and 2 are the closest to the outstanding OEIS problem, and the Borel-growth questions 3 and 4 and the growing offsets of question 8 are broader analytic projects.

Conclusion of manuscript 03, restated. The conjecture of Part I is resolved by [theorem 26.1](#), one specialization of [theorem 25.1](#), valid for every pair of directed offsets. The integer formula (28.2) proves the arithmetic phenomenon at every order, explains the delayed high-degree terms and gives efficient coefficient generation; the Borel transform and the index reconstruction are further consequences. The remaining distinction is between the stable asymptotic information, proved here with controlled remainders, and exact finite enumeration: the original guessed recurrence and a complete beyond-all-orders expansion are not claimed.

OEIS note, as manuscript 03 proposes it. Its `data/03-rational-collapse-OEIS_proposed_note.md` drafts an OEIS remark that every c_J of A189281 is an integer, with the finite formula (28.2) at $u = -1$, that the rational-collapse conjecture is proved, that the A189281 recurrence is not, and that the same holds for every directed pair including A189282–A189284. It is marked “not submitted”; it asks for a stable public citation and independent review first. Nothing was submitted.

35 Formula sheet

With $\sigma = r + s$ and $w_j = (r-1)^{\bar{j}}(s-1)^{\bar{j}}/j!$ (manuscript 03, Appendix A):

$$\begin{aligned}
\mu_k^*(n) &= \sum_{j=0}^k \frac{w_j \binom{n-\sigma+1-j}{k-j}}{n^{\underline{k+j}}}, \\
R_h^{r,s}(n) &= \sum_{j=0}^h \frac{w_j \binom{h-\sigma}{h-j}}{n^{\underline{h+j}}}, \\
B_J(u) &= \sum_{h=1}^J u^h \sum_{j=0}^{\min(h, J-h)} w_j \binom{h-\sigma}{h-j} \left\{ \begin{matrix} J-1 \\ h+j-1 \end{matrix} \right\} \quad (J \geq 1), \\
\psi_i &= \sum_{j=0}^{\lfloor i/2 \rfloor} (-1)^{i-j} w_j \binom{i-j-\sigma}{i-2j}, \\
c_J &= \sum_{i=1}^J \psi_i \left\{ \begin{matrix} J-1 \\ i-1 \end{matrix} \right\} \quad (J \geq 1), \\
\Psi(z) &= (1+z)^{\sigma-1} {}_2F_0\left(r-1, s-1; ; -\frac{z^2}{1+z}\right), \\
\widehat{C}(t) &= \mathcal{E}'(e^t - 1), \quad \mathcal{E}(\xi) = \sum_{i \geq 0} \psi_i \xi^i / i!.
\end{aligned} \tag{35.1}$$

The conventions for $J = 0$ are $B_0 = c_0 = \psi_0 = 1$. The formulas for μ_k^* and R_h are rational identities; actual moments require (3.13). Ψ is formal, while $\mathcal{E}$ and $\widehat{C}$ are entire.

The shipped files of manuscript 03. The prefix `03-rational-collapse-` marks its twelve shipped files. Its three scripts and its `Makefile` are in `code/`; its five recorded outputs, the b-file excerpt and its OEIS note are in `data/`; its `SOURCES.md` is at the report root. All are byte-identical to the delivery. `certify.py` imports its sibling by the delivered name (`from verify import ...`); `verify.py` and `inverse_demo.py` read the excerpt by its delivered name, `b189281_0_30.txt`; all three write unprefixed files into the `data/` directory beside their own; and the `Makefile` calls the scripts by their delivered names. In the shipped layout the scripts would fail to import, skip the OEIS-value checks or fail to find their input, and write beside the files of Parts I and II. The report README gives a copy-based recipe.

A Proof dependencies and archive guide

The main theorem uses the following chain, all proved in the body of the article:

$$\begin{aligned} \text{selected-edge tilings} &\implies \text{stable polynomial moments} \\ &\implies \text{uniform profile expansion} \\ &\implies \text{finite coefficient formula.} \end{aligned}$$

The separate factorial-moment bound justifies discarding the tail and also proves the total-variation universality theorem. Stirling’s expansion and the definition of Lambert W are standard external analytic inputs for the inverse section. The recurrence conjectures, the rational-collapse conjecture, and the 2026 holonomicity report are not dependencies of the all-orders theorem.

The archive contains the L^AT_EX source and compiled PDF, the core exact coefficient/enumeration library, a validation program, the rational-certificate program, two optional exploratory programs, and their outputs. The README gives commands to reproduce the computations and compile the article. The symbolic collapse script asserts only its explicitly bounded range of identities; its success is not presented as an infinite proof. [Added 1 October 2026, batch 73O2: in the shipped report the compiled PDF is a new build of this text, and the archive’s README is replaced by the report README; section B lists what is shipped.]

B Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 60 of batch 73 of ProveIt’s incoming-reports intake: the archive `oeis_path_forests.zip` (main file `article.tex`, a 21-page PDF titled *Unconditional all-orders asymptotics for two OEIS permutation sequences*). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision. It cites the repository only by the path of the Fabius completion audit [9], which it uses as an editorial example and calls “not a dependency”. This report does not continue that audit, and nothing was filed in the Fabius tree. [Added 1 October 2026, batch 74: since batch 74 the report is built from two manuscripts; Part II and the paragraph “Part II (batch 74)” below describe the second.]

What the write changed.

- Every label gained the prefix `spf`; no label was renamed otherwise and none was removed. The manuscript had 78 labels.
- Added: the editorial note after the status paragraph; Remark 7.2 on the offsets (1, 1) (A000255 and A002464); Remark 9.1 on the repository’s inversion apparatus; bracketed notes on Padhi’s preprint, on integer thresholds and on the shipped files; three bibliography entries [10, 11, 12]; and this section. A separate reciprocal note, Remark 7.4, points to the local law of the companion report on balanced Smirnov words.
- Nothing was removed. Every non-claim of the manuscript is kept: the status paragraph, the Poincaré-only reading of Table 1, the limitation after theorem 4.2, the fixed-offset limitation after the proof of theorem 1.1, the conditional nature of Proposition 11.2, and the statement that the recurrences of A189281 and A110128 are not proved.

Status of the conjecture. Conjecture 11.1 (rational collapse) is a conjecture. It has been checked as an identity of rational functions for $h = 4, \dots, 16$ only, and the integrality of every $c_j^{(1)}(2, 2)$ is conditional on it (Proposition 11.2). [Added 3 October 2026, batch 85: this paragraph describes the report before batch 85. The conjecture is proved in Part III (theorem 26.1), and the integrality is unconditional (corollary 26.3 and theorem 28.1); see the paragraph “Part III (batch 85)” below.]

Overloaded letters. The manuscript reuses a few letters; none is renamed here. C is a tiling count $C_{\mathcal{P}}(\mathbf{b})$ in sections 2 and 3, the number of non-dimer components in section 5, a constant $C_{r,s}$ in Lemma 6.1, and the series $C(x)$ in section 9. K is a moment cutoff in Lemma 4.1 and Theorem 4.2 but the edge degree of the non-dimer part of a profile in section 5. d is the defect of a profile, while $d_j(w)$ are the inverse coefficients of section 9. $\mathcal{D}$ is a formal series in section 3 and D the index correction in section 9. In each case the section determines the meaning.

Checks by the intake. The intake reran every shipped program on a copy (all passed; the regenerated outputs equal the shipped ones apart from line endings, a final newline and timings), recounted both sequences for $n \leq 9$ by brute force, compared the two order-ten expansions with the OEIS entries digit for digit, rederived d_1 in theorem 9.2 by hand, and made the (1, 1) checks of Remark 7.2. These checks do not replace an independent review of the proofs.

Part II (batch 74). [Added 1 October 2026, batch 74.] Part II is built from manuscript 03 of batch 74, the archive `ProveIt_Fixed_Gap_Asymptotics.zip` (14 files, a 25-page PDF), which arrived in commit `c664fc2f0` and was placed in commit `b669cff87`. Its pinned commit is `291fbe6bb` [14]. Its article, README, PDF and checksum ledger `SHA256SUMS.txt` (13/13 verified at placement) are not shipped; its other ten files are, byte-identical, with the prefix `02-fixed-gap-`: `code/fixed_gap.py` and `code/verify.py` in `code/`; the six files of its `data/` and `requirements.txt` in `data/`; and `sources.md` at the report root. Those files use the delivery names.

Where the batch-74 merge had to choose. Manuscript 60 stays the base, being the more general in u and the stronger in arithmetic (denominator theorem, order sixteen), in the whole-law statement and in its certificates. Manuscript 03’s re-derivations of Part I’s theorems are recorded as second routes in table 5 and not reprinted; its own results form Part II, printed after Part I’s conclusion so that no existing number changes. Its symbols are renamed into Part I’s notation (table 4); the devices its new proofs need (the moment bound with $D_{n,k}$,

the unit-disk Taylor bound, the defect-sensitive bound) are printed with them. Its coefficients through order twelve are printed only as a confirmation of [table 1](#).

What the batch-74 write changed. It added the part headings, the editorial note after the batch-7302 one, bracketed notes marked “[Added 1 October 2026, batch 74]” after the proof of [theorem 1.1](#), in questions 1 and 5 of [section 12](#) and in this appendix, Part II ([section 14](#) to [section 22](#)), two bibliography entries, and a table-of-contents line for the appendices. All its labels carry the prefix `fga:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscript 60 was changed or removed. Manuscript 03’s text is restated in Part I’s notation, with some proofs condensed; none of its statements was strengthened.

Part III (batch 85). [Added 3 October 2026, batch 85.] Part III is built from manuscript 03 of batch 85 (a different manuscript from manuscript 03 of batch 74), the archive `OEIS_Rational_Collapse_and_Integrality.zip` (16 files, a 23-page PDF), which arrived in commit `317c1ce2e` and was placed in commit `713149ded`. Its pinned commit is `d68b65ea0` [19]. Its article, README, PDF and checksum ledger `MANIFEST.sha256` (15/15 verified at placement) are not shipped; its other twelve files are, byte-identical, with the prefix `03-rational-collapse:` `code/verify.py`, `code/certify.py`, `code/inverse_demo.py` and the `Makefile` in `code/`; the six files of its `data/` and its `OEIS_proposed_note.md` in `data/`; and `SOURCES.md` at the report root. Those files use the delivery names.

Where the batch-85 merge had to choose. Manuscript 03 answers this report’s open conjecture, so it is an addition, printed as Part III after Part II so that no existing number changes. Its theorems that Parts I and II do not contain are printed in full with their proofs: the single-sum formula, the universal collapse, the A189281 product, integrality with the degree law, the kernel and auxiliary recurrence, the Borel transform, the coefficient tables beyond order 16 and for A189282–A189284, and its certificates. Its second proof of [theorem 1.1](#) at $\theta = 1$ is printed, because it runs through the new identity; its other re-derivations are recorded in [table 9](#). Its special cases that repeat Part II’s [propositions 17.2](#) and [18.1](#), which it does not cite, are printed as notes crediting Part II ([remarks 25.2](#) and [28.2](#)); its index reconstruction is printed as an instance of Part II’s [proposition 20.1](#). Its symbols are renamed into Part I’s notation ([table 8](#)), its components are indexed by vertices rather than edges, and $p = r - 1$, $q = s - 1$ are written out. Its coefficient table is printed only from order 17, orders 11 to 16 being in [table 1](#).

What the batch-85 write changed. It added two macros to the preamble, the batch-85 editorial note, bracketed notes marked “[Added 3 October 2026, batch 85]” after [Conjecture 11.1](#), after [Proposition 11.2](#), in questions 1 and 4 of [section 12](#), after [corollary 18.2](#), in the overlap paragraph and question 1 of [section 22](#), and in the paragraph “Status of the conjecture” of this appendix; Part III ([section 23](#) to [section 35](#)); these three paragraphs; and five bibliography entries. All its labels carry the prefix `spf:rc:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of either earlier manuscript was changed or removed: the conjecture keeps its heading and label, and the conditional proposition its statement. Manuscript 03’s text is restated in Part I’s notation, with some proofs condensed; none of its statements was strengthened, and every one of its non-claims is kept in [section 23](#).

References

- [1] OEIS Foundation Inc., *A189281: Number of permutations p of $1, 2, \dots, n$ satisfying $p(i+2) - p(i) \neq 2$* . <https://oeis.org/A189281>. Entry inspected 1 October 2026.
- [2] OEIS Foundation Inc., *A110128: Number of permutations p of $1, 2, \dots, n$ satisfying $|p(i+2) - p(i)| \neq 2$* . <https://oeis.org/A110128>. Entry inspected 1 October 2026.
- [3] G. Spahn and D. Zeilberger, *Counting permutations where the difference between entries located r places apart can never be s (for any given positive integers r and s)*, arXiv:2211.02550 (2022), revised manuscript with postscript. <https://sites.math.rutgers.edu/~zeilberg/mamarim/mamarimPDF/permsV2.pdf>.
- [4] R. Tauraso, *The dinner table problem: the rectangular case*, *Integers* **6** (2006), A11; arXiv:math/0507293. <https://arxiv.org/abs/math/0507293>.
- [5] M. Kauers and C. Koutschan, *Guessing with little data*, *Proceedings of ISSAC 2022*, 83–90; arXiv:2202.07966. <https://arxiv.org/abs/2202.07966>.
- [6] J. S. Padhi, *Solutions to five challenge problems in enumerative and algorithmic combinatorics, with an account of the human–machine methodology employed*, arXiv:2608.11290 (2026), especially Sections 5 and 6. <https://arxiv.org/pdf/2608.11290>. Consulted as a report of current related work; not used as a theorem dependency.
- [7] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic expansions of the gamma function*. <https://dlmf.nist.gov/5.11>. Inspected 1 October 2026.
- [8] NIST Digital Library of Mathematical Functions, *Section 4.13: Lambert W -function*. <https://dlmf.nist.gov/4.13>. Inspected 1 October 2026.
- [9] V. Reshetnikov, *ProveIt*, repository and asymptotic completion audit: `Analysis/FabiusFunction/docs/ASYMPTOTIC_COMPLETION_AUDIT.md`. <https://github.com/VladimirReshetnikov/ProveIt>. Repository material inspected 1 October 2026. Used as project context and an example of separating asymptotic assertions from their verification status.
- [10] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A000255: $a(n) = n a(n-1) + (n-1) a(n-2)$, $a(0) = a(1) = 1$* . <https://oeis.org/A000255>. Entry inspected 1 October 2026.
- [11] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A002464: Hertzprung’s problem, including the Abramson–Moser asymptotic expansion computed by V. Kotesovec*. <https://oeis.org/A002464>. Entry inspected 1 October 2026.
- [12] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, canonical volume of the ProveIt series-and-transseries collection. Its source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (labels `p0:thm:perturbed-inversion`, `p0:thm:staircase`), chapter on the increasing inverse of Gamma (`p6:thm:gamma`) and chapter “The subfactorial” (`p8:sec:top`). Repository state of placement commit `6e193dd4f`.
- [13] [Added in the batch-74 write, from manuscript 03.] R. Matsuo, V. Kotesovec and C. Koutschan, *Table of A189281 for $n = 0, \dots, 300$* , with the authorship of ranges given in the OEIS entry. <https://oeis.org/A189281/b189281.txt>. Accessed by manuscript 03 on 1 October 2026.

- [14] [Added in the batch-74 write, from manuscript 03.] V. Reshetnikov, *ProveIt*, repository and README, inspected 1 October 2026 at commit `291fbe6bb4477e0cb2131f1e041b12a9abe363fd`.
<https://github.com/VladimirReshetnikov/ProveIt>.
- [15] [Added in the batch-85 write, from manuscript 03 of batch 85.] OEIS Foundation Inc., *A189282: directed gap (3,3) avoidance*. <https://oeis.org/A189282>. Inspected by manuscript 03 on 3 October 2026.
- [16] [Added in the batch-85 write, from manuscript 03 of batch 85.] OEIS Foundation Inc., *A189283: directed gap (4,4) avoidance*. <https://oeis.org/A189283>. Inspected by manuscript 03 on 3 October 2026.
- [17] [Added in the batch-85 write, from manuscript 03 of batch 85.] OEIS Foundation Inc., *A189284: directed gap (5,5) avoidance*. <https://oeis.org/A189284>. Inspected by manuscript 03 on 3 October 2026.
- [18] [Added in the batch-85 write, from manuscript 03 of batch 85.] R. Matsuo et al., *Table of A189281 for $n = 0, \dots, 300$* ; the entry credits terms 0–35 to V. Kotesovec and 36–39 to C. Koutschan. <https://oeis.org/A189281/b189281.txt>. Inspected by manuscript 03 on 3 October 2026; it ships the excerpt $n = 0, \dots, 30$ as `data/03-rational-collapse-b189281_0_30.txt` (OEIS content, CC BY-SA 4.0).
- [19] [Added in the batch-85 write, from manuscript 03 of batch 85.] V. Reshetnikov, *ProveIt*, this report's `article.tex` (Parts I and II), inspected 3 October 2026 as blob `5b60a1ad447afe7dce58534fba447b05e401f172` at commit `d68b65ea064084fb121df9bb431b21d086f7be81`.
<https://github.com/VladimirReshetnikov/ProveIt>.